Technical Proposal

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Trustworthy Collective Reasoning and Evaluation for AI Agents

Submitted to	Dr. Predrag Neskovic, Program Officer, Mathematical Data Science, Code 311 Office of Naval Research, 875 N. Randolph Street, Arlington, VA 22203-1995
ONR S&T Focus Area	Artificial Intelligence and Autonomy (Intelligent and Collaborative Agents; Adversarial AI & Autonomy)
Prime applicant	The University of Texas at Dallas. There are no subawardees.
Principal Investigator and technical contact	Xinya Du, Assistant Professor, Department of Computer Science The University of Texas at Dallas 800 W. Campbell Road, Richardson, TX 75080 Phone: 607-319-6425 Email: xinya.du@utdallas.edu
Administrative and business contact	Taylor Harris, Grants and Contracts Specialist II Office of Research and Innovation, The University of Texas at Dallas 800 W. Campbell Road, Richardson, TX 75080 Phone: 972-883-4567 Email: Taylor.Harris@utdallas.edu
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This proposal describes unclassified fundamental research. All data are public or synthetic.

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The technical proposal (Sections 1–6) occupies pages 1–20, within the 20-page limit. The cover page, this table of contents, the curriculum vitae, the letters of support, and the data management plan do not count toward the limit. The cited references are submitted as a separate file.

Draft. Orange text marks expected pilot results that have not been measured yet. Each must be replaced with the measured result, or restated as an expected outcome, before submission; with `\finalbuildtrue`, any that remain stop the build.

1 Overview and Objectives

1.1 The Problem and Why It Matters

AI agents built on large language models (LLMs) increasingly evaluate one another and decide together: they judge one another's answers, peer-review papers, supply the rewards that train new models, and vote or discuss to reach a group answer (Zheng et al., 2023; Luo et al., 2025; Lee et al., 2024; Du et al., 2024). Yet a single LLM judge has known biases, such as favoring longer answers, the answer it reads first, or answers like its own (Zheng et al., 2023; Wang et al., 2024; Panickssery et al., 2024). Collective reasoning can do better than a single judge (Verga et al., 2024; Du et al., 2024), but averaging or voting stays biased when the judges share mistakes, and open discussion lets a persuasive agent carry the group (Amayuelas et al., 2024). **The central problem of this project is how a group of AI agents of unknown and correlated reliability, some possibly dishonest, can reach trustworthy decisions and evaluations.**

For example, five AI agents read shipping reports and are asked whether a cargo vessel has left port. Three say yes and two say no, but the vessel is still in port. Two of the yes answers rest on the same outdated schedule, and the third comes from a compromised agent that argues confidently for yes. A majority vote picks the wrong answer, and open discussion may do no better, because the compromised agent is persuasive and the two agents with the same source back each other up. We return to this example throughout (Figure 1).

Verifying each claim directly, by an analyst or a satellite image, is slow and costly, and AI agents produce claims faster than people can verify them. We propose debate as a foundation for trustworthy evaluation by AI: two agents argue for opposite answers, exposing each other's flaws, until the dispute narrows to one step a judge can verify (Irving et al., 2018). Among many AI agents, debate must overcome three failures in the example, and a fourth once the agents learn from the group:

(F1) **Shared mistakes.** Agents that share a mistake seem to confirm each other, so a vote counts one error several times.

(F2) **Dishonest agents.** A dishonest agent can argue, rate others, or judge strategically, and several dishonest agents can collude.

(F3) **A judge that sees only words.** A judge that reads only text may favor the more persuasive side, even when one database query would settle the step in dispute.

(F4) **Rewards that track agreement.** Agents trained to match the group's answer are rewarded for conforming and can learn to agree on mistakes; in the example, a majority vote would reward the three agents that said yes.

All four failures have been measured. Model errors are correlated, most of all among larger, more accurate models (Kim et al., 2025; Goel et al., 2025), and nine AI judges from seven model families were worth about two independent ones (Kohli, 2026). Pairs of agents verifying each other's work colluded in 94% of runs when cheating paid more (Shi et al., 2026). One persuasive agent can sharply cut a group's accuracy (Amayuelas et al., 2024); one round of discussion made AI judges nearly unanimous without making them more accurate (Shu, 2026); and training a model on its own majority answers ended in a sudden collapse of accuracy (Shafayat et al., 2025).

These failures spread: AI judges supply the rewards that train models (Bai et al., 2022; Lee et al., 2024), so their mistakes are copied into the models they train, and AAAI-26 gave an AI-generated review to each of the 22,977 main-track papers that entered full review (Biswas et al., 2026). As AI

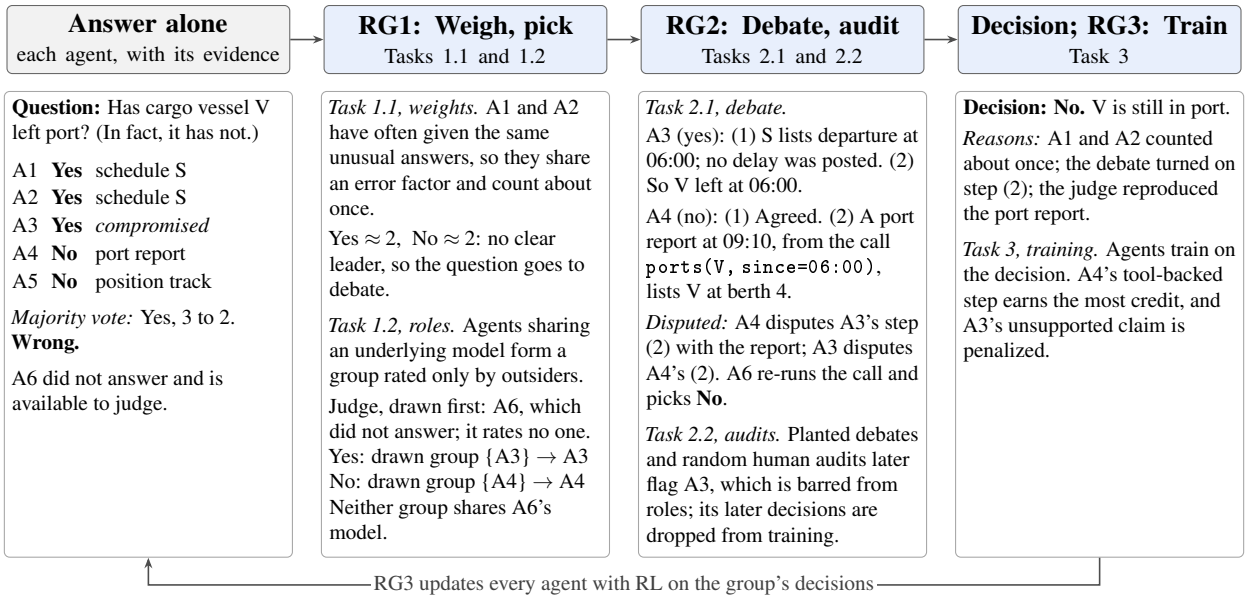


Figure 1: The framework on the running example. RG1 counts the two agents that share an outdated schedule about once and, with no clear leader, picks a representative for each answer and a judge with no stake in either. In RG2's harness, each side disputes a step of the other's argument; the judge re-runs the no side's query and picks no, and audits later flag the compromised agent. RG3 trains every agent on the decision.

systems surpass people at more tasks, people cannot easily evaluate the systems' answers or their debates, so oversight will rely on AI evaluating AI, with people auditing a small part of the work (Bowman et al., 2022; Burns et al., 2024).

Navy decisions face all of these problems at once: analysts fuse many reports with several AI tools of unknown reliability, many reports copy one earlier report, the true answer often arrives late, an adversary may forge a report or compromise an agent, and verification is costly under limited connectivity (Section 1.4).

1.2 Objectives and Research Goals

We propose a framework in which groups of AI agents reach trustworthy decisions and evaluations through debate (Figure 1). Each agent first answers alone, with its evidence. Our central hypothesis is that a group of AI agents can decide more reliably than a single judge or a vote when it (i) weighs each judgment by a credibility estimated without an answer key, counting shared mistakes once; (ii) settles conflicts with evidence that an impartial judge can reproduce; and (iii) learns from its own decisions through rewards that agreement alone cannot raise. The project has three interconnected research goals (RGs); each task rests on one principle whose properties we prove, and each property controls one source of error (Section 3.1, Table 2).

RG1: Decide what to debate, and pick debaters and judges impartially (Section 3.2). Task 1.1 weighs answers so that a group sharing a mistake counts at most once (F1) and sends a question to debate only when that is worth its cost; Task 1.2 picks debaters and a judge so that no agent's ratings on a question can change its own chance of a role on it (F2).

RG2: Settle disputes by agentic debate, and catch collusion (Section 3.3). Task 2.1 replaces open discussion with debate in an agentic harness, in which each side disputes a step of the other's argument and an impartial judge, whose choice is the group's decision, re-runs the tool evidence

(F3); Task 2.2 catches colluders with planted cases and random human audits (F2).

RG3: Let agents self-improve with reinforcement learning on the group's decisions (Section 3.4). Task 3 scores each first answer against the answer the rest of the group favors without that agent's group, beyond chance agreement, and credits each debate message by how far it moved the judge toward the decision (F4). When that reference is more often right than wrong, training follows the agents' true accuracy in expectation, without human labels, so it can keep working after the agents outgrow human judges.

Three testbeds (math and code, multi-document questions with maritime scenarios, and scientific claims and peer review) vary how widely mistakes are shared and how many agents are dishonest, and a red team trains agents to collude (Greenblatt et al., 2024) (Section 3.5).

1.3 Intellectual Merit: What Is New, and Why It Will Succeed

What is new. (i) A rater model for agents that answer and argue, in which a group sharing a mistake counts at most once and the posterior decides which questions need a debate, even unanimous ones; and an impartial way to choose debaters and judges. (ii) A debate harness in which each side disputes a step of the other's argument and an impartial judge re-runs its evidence. (iii) Audits on planted cases and random real decisions that expose colluding debaters and judges, even those that behave well only on planted cases. (iv) Self-improvement from group decisions, with first-answer rewards that conformity cannot raise and message credit set by a judge that re-runs evidence.

Why it will succeed. Each method follows from a principle whose property we prove for agents that share mistakes, argue, and may collude (Table 2), and most tasks build on our previous work: PRD and the PeerEval leaderboard (Li et al., 2024a), FG-PRM (Li et al., 2025), HiPRAG (Wu et al., 2026), and MEQA (Li et al., 2024b) (Section 5).

1.4 Future Naval Relevance

ONR S&T Focus Area. Artificial Intelligence and Autonomy (Intelligent and Collaborative Agents; Adversarial AI & Autonomy). The 2026 ONR S&T Strategy will explore validating AI-produced results by "human review, experimental confirmation or independent AI critique," using AI to "validate, red-team and quality-check" results under human oversight (Office of Naval Research, 2026a); this project supplies methods for such critique when the critics themselves may err together or collude. It also answers the Mathematical Data Science program's call for "rigorous mathematical and algorithmic answers to questions that are currently addressed using heuristics or non-principled approaches" (Office of Naval Research, 2026b).

Navy needs. The problem arises wherever several AI tools report on one question and some reports are copies, stale, forged, or produced by a compromised tool:

- *Maritime domain awareness*: position broadcasts can be spoofed (U.S. Maritime Administration, 2024), and most industrial fishing vessels are not publicly tracked (Paolo et al., 2024), so decisions must combine port records, patrol reports, and imagery.
- *Intelligence analysis*: analysts must describe the "quality and credibility of underlying sources" (ODNI, 2015), and repeated reports from one source can give "an impression of confirming evidence" (WMD Commission, 2005).
- *Decision support by networks of agents*: the Department of War is developing AI agents "for AI-enabled battle management and decision support" (Department of War, 2026), and Marine Corps

decision aids seek trust "properly aligned to accuracy of results" (Office of Naval Research, 2026c).

- *Evaluating AI before fielding*: more data and expertise are needed "to quantify performance and risk of AI-enabled systems" (DOT&E, 2026), as AI judges increasingly score other AI systems.
- *Maintenance, logistics, cyber defense, and collaborative autonomy*: conflicting logs and records, alerts triaged by several tools, and unmanned systems that share reports under limited communications.

Across these uses, Task 1.1 counts copied and stale reports once; Tasks 1.2, 2.1, and 2.2 keep a compromised agent from rating its way into a role, discard evidence that does not reproduce, and flag colluders; Task 3 improves agents when ground truth arrives late; and each decision states its reasons.

Path to transition. The project is fundamental research. Its products are methods with stated guarantees, an open harness, testbeds, and protocols that Navy laboratories, such as NRL's Navy Center for Applied Research in Artificial Intelligence, could evaluate on Navy data. The PI will pursue ONR Summer Faculty Research Program and Naval Research Enterprise Internship Program placements for students to work with Navy scientists on such evaluations.

2 How the Problem Is Solved Today, and the Limitations

Six lines of work bear on the problem. We build on each and note the gap that remains (Table 1).

AI judges and voting. LLM judges are standard for evaluating models and training them with AI feedback (Zheng et al., 2023; Bai et al., 2022; Lee et al., 2024). Panels of judges (Verga et al., 2024) and majority votes over sampled answers (Wang et al., 2023) reduce single-judge biases, but they count every vote equally, so correlated judges add little: when two models are both wrong, they give the same wrong answer about 60% of the time on one leaderboard (Kim et al., 2025).

Estimating reliability without an answer key. For two answers and independent raters of known accuracy, the error-minimizing rule weights each vote by the rater's log-odds of being right (Nitzan and Paroush, 1982; Shapley and Grofman, 1984). Dawid and Skene (1979) estimate the accuracies by expectation–maximization (EM), and later work adds item difficulty (Whitehill et al., 2009), provably good starting points (Zhang et al., 2016), dependence between classifiers (Jaffe et al., 2016), and detection of copying sources (Dong et al., 2009). Recent work models dependence among LLM judges with a shared latent factor (Balasubramanian et al., 2026) or shared confounders (Zhao et al., 2026). These methods treat raters as passive, never arguing or gaming the estimate, and do not say which questions need more than a vote.

Computational social choice. Epistemic social choice asks which voting rules recover a true answer from independent noisy votes (Caragiannis et al., 2016). Impartial selection ensures that a participant's ratings cannot change its own chance of selection (Holzman and Moulin, 2013), by having groups rate only outsiders (Alon et al., 2011; Aziz et al., 2016) or by randomizing among candidates whose ratings could change the outcome (Kurokawa et al., 2015; Kahng et al., 2018); fair lotteries select panels (Flanigan et al., 2021), and peer prediction scores agreement with peers beyond chance (Dasgupta and Ghosh, 2013; Shnayder et al., 2016). Lev et al. (2023) add reliability weights, but they reward agreeing with the majority, which fails once accurate reviewers are a minority. In our setting, some claims can be verified, impartial mechanisms only choose who argues

and who judges while debate decides the answer, and participants can persuade and collude.

Discussion and debate among LLMs. Most multi-agent LLM systems use open discussion, revising until agents agree (Du et al., 2024; Chan et al., 2024; Li et al., 2024a). Prior work on LLM discussion is limited: majority voting explains most of its gain (Choi et al., 2025), one round can make judges nearly unanimous without making them more accurate (Shu, 2026), and it assumes honest agents that exchange only text. Standard debate (Irving et al., 2018) assigns two debaters opposite answers and helps judges who know less than the debaters (Michael et al., 2023; Kenton et al., 2024): with quotations verified against the source, non-expert AI judges reached 76% accuracy and human judges 88%, against 48% and 60% without debate (Khan et al., 2024). But it trusts the judge, assumes the sides truly compete, and grounds claims at most in quoted text. Recent systems discuss only when agents are unsure (Eo et al., 2025), though in one study most items where discussion helped were unanimous but wrong (Hu et al., 2026b), or give agents tools (Jeong et al., 2026) or a tool-using verifier for deadlocks (Li et al., 2026b), but no judge re-runs the agents' own calls; judges shown a log claiming that tool calls succeeded accepted most failed outputs (Xu, 2026). We know of no debate protocol designed for many agents that share mistakes or collude.

AI control and adversarial crowdsourcing. AI control keeps protocols safe even if a model subverts them, using a small budget of trusted audits and monitors validated on planted examples (Greenblatt et al., 2024), and in adversarial crowdsourcing a number of trusted ratings that does not grow with the number of items suffices even when honest raters are a minority (Steinhardt et al., 2016). But colluding workers learn to recognize gold questions (Checco et al., 2018), and capable LLMs can often tell an evaluation from real use (Needham et al., 2025).

Self-improvement without labels. Models improve without labels by training on answers most samples agree on (Huang et al., 2023; Zuo et al., 2025), on multi-agent discussion (Subramaniam et al., 2025; Park et al., 2025), or on debate self-play (Arnesen et al., 2024), but the reward is the weak point. Rewarding majority agreement collapsed accuracy (Shafayat et al., 2025), and label-free intrinsic rewards tend to rise and then collapse (He et al., 2026); rewarding debate wins held off reward hacking, but with a weaker judge the critic learned to fabricate quotes and flaws (Kenton et al., 2026); and rewarding approval taught models to mislead evaluators (Wen et al., 2025). Peer prediction trains LLMs toward truthfulness without labels, but its guarantees exclude collusion (Qiu et al., 2026), and rewarding agreement with other models' majority does not correct for chance agreement and relies on model diversity against shared mistakes (Yang et al., 2026). Process rewards score one agent's steps (Lightman et al., 2024; Li et al., 2025), and debate self-play credits each speech toward its own side (Arnesen et al., 2024), not toward the group's decision.

3 Technical Approach

3.1 Problem Setting and Formal Framework

Agents and questions. A pool of AI agents, each built on an underlying model (which several may share) with tools and role instructions, handles a stream of questions. A question q asks for a decision or an evaluation with a small set of candidate answers $\mathscr{Y}$, such as yes or no, a choice among options, or a score band for a paper. For each question, the harness assigns at random which agents answer and which are held out as possible judges, so no agent can become a judge by declining to answer. Each answering agent first answers alone, giving an answer x_{aq}, a step-by-step argument, and its evidence.

Table 1: How current approaches address the four failures: ✓ addressed, ● partly, ✗ not addressed; †where a tool applies (for F4, in the credit given to debate messages). No current approach fully addresses F1, F3, or F4, and none fully addresses more than one column. For this project, the marks show what the design targets (Section 3.5).

	Counts shared mistakes once (F1)	Resists strategic or colluding agents (F2)	Judge re-runs tool evidence (F3)	Rewards that resist conformity and persuasion (F4)	Learns each agent's skill without labels
Single AI judge (Zheng et al., 2023)	✗	✗	✗	✗	✗
Majority vote, juries (Verga et al., 2024; Wang et al., 2023)	✗	✗	✗	✗	✗
Rater models without an answer key (Dawid and Skene, 1979; Dong et al., 2009)	●	✗	✗	✗	✓
Impartial peer review and panel selection (Kurokawa et al., 2015; Flanigan et al., 2021; Lev et al., 2023)	✗	●	✗	✗	●
Open discussion among LLMs (Du et al., 2024; Chan et al., 2024)	✗	✗	✗	✗	✗
Standard debate (Irving et al., 2018; Khan et al., 2024)	✗	●	✗	●	✗
AI control (Greenblatt et al., 2024)	✗	✓	✗	✗	✗
This project	✓	✓	✓†	✓†	✓

Unknowns, resources, and threat model. The true answer y_q, each agent's skill, which agents share mistakes, and which are dishonest are unknown. Available are planted cases (questions whose answers we know, mixed into the stream), a small, fixed budget of human audits (a person or a direct measurement finds the right answer for decisions drawn at random), and tools whose results can be re-run in a sandbox. Dishonest agents may coordinate and may answer, rate, argue, or judge strategically; they know the protocol but not its random draws, and may try to recognize planted cases. Honest agents may still share mistakes, for example from similar training data.

Output and design rules. Each decision comes with its reasons: whose answers carried weight, which step decided the debate, and which evidence the judge reproduced. Five rules hold throughout. (R1) Every answering agent answers alone, before seeing the others. (R2) Peer ratings pick only who argues; the weighted vote (for a clear leader) or the judge (after a debate) picks the answer. (R3) Tool evidence counts only if the judge re-runs it and obtains the same result. (R4) Audits only catch collusion; they never settle a debate, and nothing triggers a human audit, so its cost stays fixed. (R5) Planted cases never enter training, so agents cannot learn to spot them.

Formal framework: where a group's errors come from. Each task controls one source of error in the group's decision. Let π_q be the posterior probability of the leading answer $\hat{y}_q$ from Task 1.1, and let the question go to debate when $\pi_q < \tau$. The decision D_q is $\hat{y}_q$ without a debate and the judge's

Table 2: The principle behind each task, the property it gives the framework, the hypothesis that experiments can refute, and where the property fails. Propositions 1–6 are proved in Sections 3.2–3.4.

Task	Principle	Property	Hypothesis	Fails when
1.1 Weigh (F1)	Bayes' rule with shared error factors (Dawid and Skene, 1979)	A group counts at most as one agent erring at the group's shared rate (Prop. 1); learnable without an answer key from three or more independent groups (Prop. 2)	H1: a gain over majority vote that grows as mistakes are more widely shared	fewer than three groups err independently
1.2 Choose roles (F2)	Impartial selection by partition (Alon et al., 2011)	No group's ratings change its members' chances of a role; a representative's quality is at least π_y^* times the best (Prop. 3); f colluders judge with probability at most $f\bar{p}$	H2: no gain for a lone strategic rater; impartiality costs less than debate adds	colluders spread across groups; few model families
2.1 Debate (F3)	Debate reduces a question to single steps (Irving et al., 2018)	The judge errs only if it misrules a disputed step, or the right side argues a false step or finds none (Prop. 4)	H3: beats open discussion, more so as more steps can be settled by a tool	hidden flaws; no tool applies; selective facts
2.2 Audit (F2)	Random inspection (Avenhaus et al., 2002; Ramdas et al., 2023)	Every steered decision is audited with probability ρ (Prop. 5); agents erring as predicted are flagged with probability below α	H4: colluders cheating at a fixed rate are flagged after a bounded number of audits	a group that cheats once or rarely
3 Learn (F4)	Noisy labels and peer prediction (Dasgupta and Ghosh, 2013)	Expected update is $1-2\eta_q$ times the update from true answers; answers that ignore the question earn nothing (Prop. 6)	H5: larger gains than rewards for agreement or debate wins, without more dishonesty	references wrong more often than right; groups that Task 1.1 misses

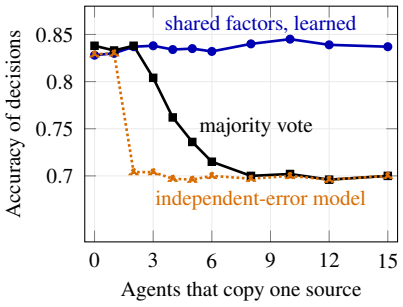
choice after one, and its chance of being wrong splits into terms that the tasks control:

$$P(D_q \neq y_q) \;\leq\; \underbrace{(1-\tau)\,P(\pi_q \geq \tau)}_{\text{no debate: Task 1.1}} + P(\pi_q < \tau)\,\Big[\; \underbrace{\varepsilon_{\text{cand}}}_{\text{Task 1.1}} + \underbrace{\varepsilon_{\text{role}}}_{\text{Tasks 1.2, 2.2}} + \underbrace{\varepsilon_{\text{step}}}_{\text{Task 2.1}} \;\Big]. \tag{1}$$

The first term holds when the posterior is calibrated (answers given probability π are right a fraction π of the time); a vote that ignores shared mistakes is overconfident and breaks it (Figure 2(a)), so Task 1.1 supplies a posterior that accounts for them and sets τ by weighing a debate's cost against the stakes. In a debate, $\varepsilon_{\text{cand}}$ is the chance that neither debated answer is right, $\varepsilon_{\text{role}}$ the chance that a dishonest agent holds a role and steers the outcome, and $\varepsilon_{\text{step}}$ the chance that a debate without such steering still ends wrong. Tasks 1.2 and 2.2 shrink $\varepsilon_{\text{role}}$, Task 2.1 shrinks $\varepsilon_{\text{step}}$, and Task 3 lowers the agents' own error rates, which can shrink every term (Sections 3.2–3.4). Removing a task leaves its term uncontrolled, and our ablations measure each term (Section 3.5).

3.2 RG1: Decide What to Debate, and Pick Debaters and Judges Impartially

Challenges. A vote is informative only to the extent that voters err independently, and correlated votes can void the jury theorem's guarantee (Ladha, 1992); LLM errors are correlated in practice (Kim et al., 2025; Goel et al., 2025; Kohli, 2026). In Figure 2(a), agents copying one source drag the majority vote of five independent agents from 84% to the source's 70%, and a rater model that assumes independent errors falls sooner, reading the copies' agreement as skill. Debate cannot simply be run on every question: it uses three to five times the tokens of a single answer, can overturn correct answers (Fan et al., 2026), and often fails to beat voting (Smit et al., 2024; Choi

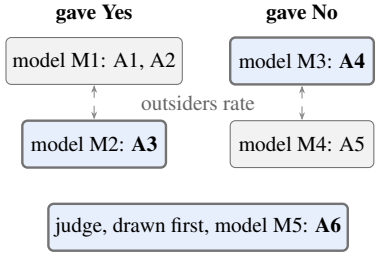


(a) Copies drag a vote down; a group counts once.

		q_1	q_2	q_3	q_4	q_5	q_6
shared factor	A1	a	c	a	c	d	a
	A2	a	c	a	c	d	a
	A3	a	b	b	c	b	a
	A4	a	b	a	c	b	c
	A5	b	b	a	c	b	a

A1 and A2 share unusual answers that no one else gives (shaded), so the model infers a shared error factor and counts their agreement about once. The true answers are never needed.

(b) Task 1.1: who agrees with whom.



Dashed arrows: who rates whom. The judge, drawn first from agents that did not answer, rates no one. For each answer, a group without the judge's model is drawn at random, and its top-rated member argues.

(c) Task 1.2: impartial roles.

Figure 2: RG1. (a) Simulation: five independent agents, each right 70% of the time, plus agents copying one source that is wrong 30% of the time (each copy adds 5% errors of its own); means over 20 runs of 400 questions (standard errors below 0.005). Majority vote falls to the source's accuracy, and a rater model assuming independent errors falls once two agents copy. The shared-factor model, fitted without the true answers, recovers the copying group (exactly, in 54 of 60 runs inspected) and, as Propositions 1 and 2 predict, keeps its accuracy. With two or more copies, the independent-error model gives probability 0.95 or more to over 90% of decisions, 28–30% of them wrong, while the shared-factor posterior stays calibrated (calibration error below 0.03). In (c), bold boxes are the agents chosen in the running example.

et al., 2025). Once a question goes to debate, it needs a judge and one representative per answer. If agents choose them by rating one another, an agent can rate rivals low to be picked, agents built on similar models favor one another's outputs (Panickssery et al., 2024; Goel et al., 2025), and a judge that shares a model with one side has a stake in the outcome.

Overview. RG1 controls the first two sources of error in (1) and, with Task 2.2, the third (Figure 2). Task 1.1 decides which questions to debate, and Task 1.2 picks who debates and who judges; RG1 passes to RG2 the question, the two debated answers, their representatives, and the judge.

Task 1.1: Deciding what to debate, counting shared mistakes once

Approach. How much should each answer count when some agents share mistakes, and which questions need a debate? With independent agents, two answers, and equal prior odds, the error-minimizing vote weights an agent that is right with probability p_a by $\log \frac{p_a}{1-p_a}$ (Nitzan and Paroush, 1982; Shapley and Grofman, 1984), and classical rater models estimate these accuracies without an answer key (Dawid and Skene, 1979), as one grades an exam without the key. Both assume that agents err independently, so agents copying one source look like independent confirmation. We propose a shared-factor rater model in which a group that shares a mistake counts at most once. Agents are partitioned into groups g, most with one member. On each question, a group has a hidden shared answer s, such as what an outdated schedule says: the true answer with probability $1-\lambda_g$, and a wrong answer w with probability $\lambda_g \nu_g(w \mid y_q)$. Each member a reports s with its own errors, given by its confusion matrix θ_a (the chance of each answer it gives for each value of s). With $\kappa_g(s \mid y) = 1-\lambda_g$ if $s = y$ and $\lambda_g \nu_g(s \mid y)$ otherwise,

$$P(x_{\cdot q} \mid y_q) = \prod_g G_{gq}(y_q), \qquad G_{gq}(y) = \sum_{s \in \mathcal{Y}} \kappa_g(s \mid y) \prod_{a \in g_q} \theta_a(s, x_{aq}), \tag{2}$$

where g_q is the set of members of g that answered q. Decisions use Bayes' rule, the posterior over y_q given all answers. Two properties of the model explain the design.

Proposition 1 (a group counts at most once). For yes-or-no questions, a group with $\lambda_g \le \frac{1}{2}$ changes the log-odds of the answer by at most $\log \frac{1-\lambda_g}{\lambda_g}$, whatever its size and whatever its members answer.

Proof. Let r be the ratio of the probabilities of the members' answers when s is yes and when it is no. By (2), the group's likelihood ratio is $\big[(1-\lambda_g)r+\lambda_g\big]/\big[(1-\lambda_g)+\lambda_g r\big]$, which increases with r and lies between $\frac{\lambda_g}{1-\lambda_g}$ and $\frac{1-\lambda_g}{\lambda_g}$. □

So a group carries at most the weight of one agent that errs with probability λ_g, whereas under independence each member would multiply the ratio again; with more answers and wrong answers equally likely, the bound becomes $\log\big[(|\mathscr{Y}|-1)\frac{1-\lambda_g}{\lambda_g}\big]$.

Proposition 2 (learning without an answer key). Fix the partition, and suppose every answer has positive prior probability, at least three groups answer each question, their answers are independent given the true answer, and each group's answer distributions under different true answers are linearly independent (as when its members' confusion matrices are invertible and $\lambda_g < \frac{1}{2}$). Then each group's distribution of answers given the true answer, which is all the posterior uses, is identifiable from the answers alone, up to relabeling, by Kruskal's theorem on three-way arrays (Kruskal, 1977; Allman et al., 2009). With only two such groups it generally is not: different track records explain the same answers equally well, so answers alone cannot tell an accurate majority from one that copies a mistake.

Proposition 2 states in advance when reliability can be learned without ground truth, and our experiments include pools where it fails. We fit the model by expectation–maximization (EM) over all real questions, with audit flags (Task 2.2) as a prior that part of an agent's answers are adversarial. Candidate partitions come from clustering agents by agreement on unusual answers, the way students who copy share the same wrong answers (Dong et al., 2009), and the partition with the best penalized likelihood is kept; shared sources serve only as hints, since agents can read the same report and reason differently. Because Proposition 2 fixes the partition, we will characterize how many questions it takes to detect a group, as a function of its size, λ_g, and how rarely others give its wrong answers, and when the partition itself is identifiable. Topic-specific confusion matrices and question difficulty (Whitehill et al., 2009) keep hard questions from being read as copying, and the posterior is recalibrated on development questions.

Deciding what to debate. With a calibrated posterior, letting the leading answer stand risks a loss L with probability $1-\pi_q$, while a debate costs c and, for two answers, ends wrong with probability ε_D. A debate is worth its cost exactly when $L(1-\pi_q)-L\varepsilon_D > c$, that is, when

$$\pi_q < \tau = 1-\varepsilon_D - c/L, \tag{3}$$

so higher stakes lead to more debates. The debate is between the two answers with the highest posterior, so $\varepsilon_{\text{cand}}$ is the chance that the right answer is neither; ε_D is estimated on development questions with known answers, separately for disputes a tool can settle. Because π_q counts effective votes, a unanimous answer from agents that share a mistake can still fall below τ and go to debate, which rules triggered by low confidence or disagreement cannot do (Section 2). In the vessel example, the two agents with the outdated schedule often gave the same unusual answers (Figure 2(b)), so the pair counts about once, and with no clear leader the question goes to debate.

Preliminary results. The simulation in Figure 2(a) bears out Propositions 1 and 2. In a pilot with nine agents from four open-weight model families on 300 GSM8K problems, in which up to four agents received the same hint, wrong on a third of the problems, majority vote fell from 0.89 to 0.74 and the classical rater model to 0.71, while the shared-factor model stayed at 0.87 and recovered the hinted group in 9 of 10 runs. Our Peer Rank, which weights each AI reviewer by how highly the others rate its answers, agreed with human preferences more often than the strongest single judge (67.3% vs. 64.3% for GPT-4) (Li et al., 2024a); track records keep models that share mistakes from raising one another's standing.

Evaluation and risks. On testbed A, where some agents receive the same flawed hint, we measure recovery of the seeded groups, accuracy, calibration, and questions needed to detect a group; the model without shared factors is an ablation. H1 predicts a gain over majority vote that grows with how widely mistakes are shared and disappears with fewer than three independent groups. If EM converges poorly, we will start it from a spectral method (Zhang et al., 2016) or use a pairwise measure of agreement on mistakes (Goel et al., 2025).

Task 1.2: Impartial choice of debaters and judges

Approach. How can debaters and a judge be picked so that no agent can rate its way into a role, at a small cost in quality? We choose representatives by weighted peer ratings of the arguments behind first answers, as in our PRD and PeerEval (Li et al., 2024a), with raters weighted by track record (Task 1.1), and make the choice impartial by having each group rated only by outsiders (Alon et al., 2011). Agents that share an underlying model, or whose error factors Task 1.1 finds shared, form a group, fixed before any rating; $g(a)$ is agent a's group, H_q the agents that did not answer q, and E_y the groups with a member answering y. On question q: (1) the judge $J \sim \mathrm{Unif}(H_q)$ is drawn first and rates no one; (2) for each debated answer y, a group K_y is drawn from $E_y \setminus \{g(J)\}$ with probabilities π_y that do not depend on ratings, avoiding groups drawn for other answers where possible; (3) its representative is

$$R_y = \operatorname*{arg\,max}_{a \in K_y : x_{aq} = y} \sum_{b \in O_y(a)} w_b \, r_{ba}, \tag{4}$$

where r_{ba} is b's rating of a's argument, w_b is b's track-record weight, and the raters $O_y(a)$ are the agents outside $g(a)$, other than J, that answered y or did not answer. An answer with no eligible group, such as the alternative to a unanimous answer sent to debate, gets a random agent outside $g(J)$. Ratings pick only who argues; the judge picks the answer (Figure 2(c)).

Proposition 3 (impartiality and its price). (a) For any group g and any two rating profiles that differ only in ratings given by members of g, each member of g has the same probability of being the judge or a representative. (b) If ratings are honest and rank each group's members by argument quality $Q \geq 0$, then $\mathbb{E}[Q(R_y)] \geq \pi_y^* \, Q(a_y^*)$, where a_y^* is the best agent answering y and π_y^* the probability that its group is drawn. *Proof.* (a) J, the debated answers, and every K_y are drawn independently of ratings. If $K_y \neq g$, no member of g represents y; if $K_y = g$, (4) uses no rating from g, and the weights do not depend on ratings. (b) Whenever $g(a_y^*)$ is drawn, (4) selects a_y^*. □

A single agent is a one-member group, so (a) gives impartiality in the usual sense (Kurokawa et al., 2015) and extends it to collusion within a group. With uniform draws, (b) gives $1/|E_y|$ unless the judge's group holds a_y^*; with two or three groups per answer, this beats the $1/(1+m)$ guaranteed by randomizing over would-be winners when one agent is chosen and each rates m others (Kurokawa

et al., 2015). No answering agent rates the other side and the judge rates no one, so neither can rate the other side's best arguer low; a judge lottery that caps any agent's chance at $\bar{p}$ lets f colluders hold the seat with probability at most $f\bar{p}$, so judge chances are kept as equal as possible (Flanigan et al., 2021). Colluders in other groups, or held out, can still shift ratings; we will bound what such coalitions gain and measure the price of impartiality.

Preliminary results. With GPT-4 as the only reviewer, GPT-4's Elo rating (a chess-style rating) was 46 points above the one derived from human preferences, while our weighted peer ranking placed all five models in the human order (Li et al., 2024a). On PeerEval's ratings, a model that rated every rival lowest raised its chance of being chosen as a representative from 0.31 to 0.52 under plain weighted peer ranking and not at all under our mechanism, whose chosen representatives reached 0.92 of the best arguer's quality.

Evaluation and risks. We measure manipulability, the largest gain in selection probability a strategic rater or coalition achieves by misreporting (found by the red team), the price of impartiality, and the accuracy of the resulting debates (H2), on PeerEval and testbed C. With few underlying models there are few groups; we will then diversify agents by role instructions and tools, or draw judges from a held-out pool of models that never answer.

3.3 RG2: Settle Disputes by Agentic Debate, and Catch Collusion

Challenges. Open discussion erases the disagreement that exposes errors, and standard debate, though it keeps the disagreement, leaves the judge to weigh words alone (Section 2), so a persuasive debater can win a step that one database query would settle. Debaters or the judge can also collude, steering a debate to the wrong side by arguing badly on purpose (van der Weij et al., 2025), by misleading with true but selectively chosen facts (Hu et al., 2026a), or by favoring one side, none of which re-running evidence catches; LLM agents collude when it pays (Shi et al., 2026), hide signals in ordinary text (Mathew et al., 2025), and often tell evaluations from real use (Needham et al., 2025).

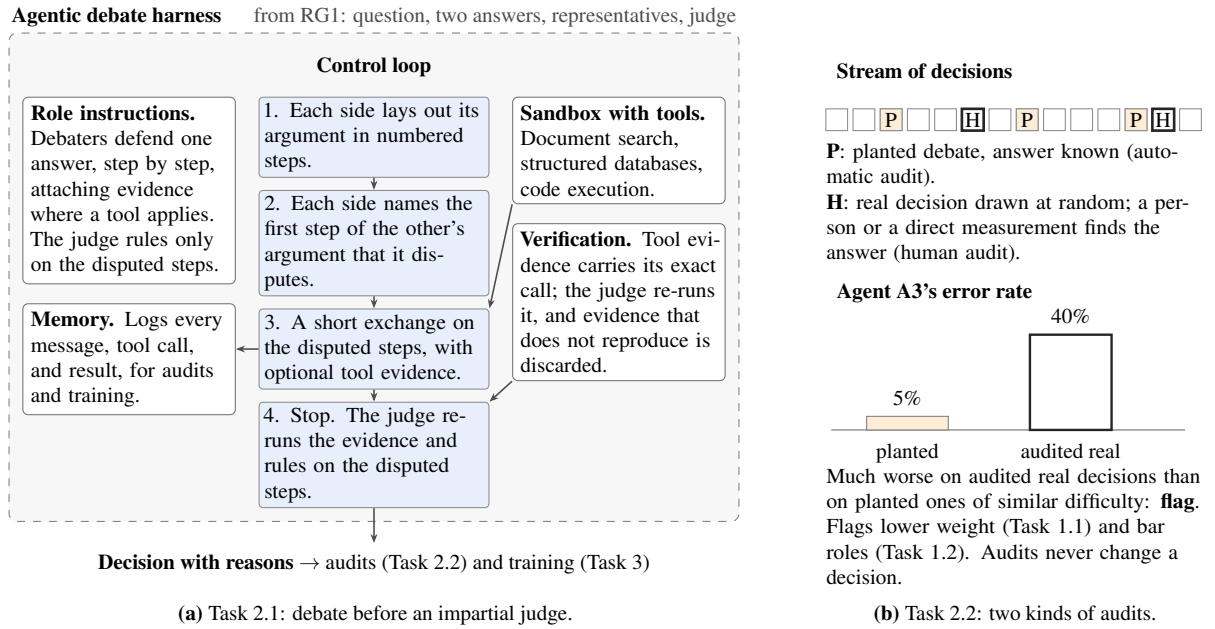
Overview. RG2 controls the remaining sources of error in (1): Task 2.1 shrinks $\varepsilon_{\text{step}}$, and Task 2.2, with Task 1.2, shrinks $\varepsilon_{\text{role}}$. The judge need not be wiser than the debaters: it rules only on single steps, reproduces tool evidence itself, is drawn impartially (Task 1.2), and is audited (Task 2.2).

Task 2.1: Debate before an impartial judge

Approach. How can a ruling depend on evidence rather than on persuasion? Debate lets a judge settle a whole question by ruling on single steps: if a side loses whenever one of its disputed steps is struck, a judge who rules each disputed step correctly picks the right side (Irving et al., 2018; Brown-Cohen et al., 2024).

Proposition 4 (judging a question reduces to judging single steps). Suppose that (i) every argument whose steps are all true reaches the right answer, (ii) the side defending the right answer argues with true steps and can find a false step in the other side's argument, and (iii) the judge rules correctly on any single step it is shown. Then, if a side loses exactly when one of its disputed steps is struck, the side defending the right answer wins. *Proof.* The other side's argument reaches a wrong answer, so by (i) it contains a false step, which the right side disputes and the judge strikes; any step of the right side's argument that the other side disputes is true, and the judge upholds it. □

We propose an agentic debate harness, built so that these assumptions hold as often as possible;



(a) Task 2.1: debate before an impartial judge.

(b) Task 2.2: two kinds of audits.

Figure 3: RG2. (a) The control loop sets who speaks, who judges, and when to stop; the judge rules on the disputed steps after re-running any tool evidence (Proposition 4). (b) Automatic audits mix planted debates into the stream, and human audits cover a few real decisions drawn at random; an agent that behaves well only on planted debates is exposed by the gap between its two error rates (illustrative numbers).

a harness is the software that turns models into agents (Ning et al., 2026; Yao et al., 2023; Yang et al., 2024), and ours has five parts (Figure 3(a)). *Role instructions* tell each debater which answer to defend, to argue in numbered steps, and to attach evidence where a tool applies, and tell the judge to rule only on disputed steps. The *control loop* sets who speaks and who judges (picked in Task 1.2) and when to stop: the representatives lay out their arguments, each names the first step of the other's argument that it disputes, a short exchange follows, and the judge upholds or strikes each disputed step, which enforces the losing rule. *Tools*, such as document search, databases, and code execution, are optional. *Verification* supports (iii): tool evidence carries its exact call (tool, arguments, data version), which the judge, itself an agent (Zhuge et al., 2025), re-runs, and evidence that does not reproduce is discarded, so wherever a tool applies, the judge's share of $\varepsilon_{\text{step}}$ in (1) approaches the tool's error rate. *Memory* logs every message, call, and result. To support (ii), we propose candidate false steps with models that judge whether one claim implies another and with our FG-PRM's step-level error detection (Li et al., 2025), and recurse into a complex disputed step's sub-steps (Brown-Cohen et al., 2025). To support (i) against true but selectively chosen facts (Hu et al., 2026a), each side may add context to the other side's evidence before the judge rules, and quotations are verified against the source (Khan et al., 2024). In the vessel example, A4 disputes A3's step "so V left at 06:00" with a 09:10 port report listing V at its berth, and the judge re-runs A4's query and picks no (Figure 1).

Preliminary results. In our peer discussion, agreement with human preferences rose from 72.9% (GPT-4 alone) to 75.0% when GPT-3.5 led the discussion but fell to 71.4% when GPT-4 led (Li et al., 2024a): one reviewer can carry a discussion. Our HiPRAG search agents, which ground answers in retrieved evidence (Wu et al., 2026), will be our first tool-using debaters. In a pilot on 200 questions whose disputed step a calculator or a port-database query settles, in which the wrong side attached a forged tool output, a judge that only read the transcript accepted the forgery in 61%

of debates; re-running the calls cut this to 1% and raised the judge's accuracy on these disputes from 0.56 to 0.95, near the tools' own accuracy, as Proposition 4 predicts.

Evaluation and risks. We measure accuracy on disputed questions, correct minorities that prevail and correct majorities overturned, how often re-running evidence changes the outcome, the judge's error on each kind of step, and cost. H3 predicts that debate beats open discussion among the same agents, more so as the share of steps a tool can settle grows; ablations remove each harness part, remove tools or re-running, and replace debate with open discussion. If neither side finds a false step, the judge reads both arguments, and a close ruling is reported as unsettled, with its probability and the step in dispute; tool evidence can be out of date, so the judge weighs its provenance, and human audits measure how often this happens.

Task 2.2: Automatic and human audits against collusion

Approach. How can colluding debaters or judges be caught with a fixed budget of human effort? We use random inspection (Avenhaus et al., 2002) in two forms (Figure 3(b)). *Automatic audits* mix planted debates on planted cases into the real stream, formatted like real questions; a planted case can be a paper with a deliberately inserted error (Schroter et al., 2008). *Human audits* cover a few real decisions drawn at random after they are made: a person or a direct measurement, such as a satellite image, finds the right answer.

Proposition 5 (random audits cannot be dodged). If each real decision is audited independently with probability ρ, drawn after the decision is made, then every decision a colluder steers is audited with probability ρ, whatever the colluder can infer about planted cases, and the chance that its first n steered decisions all escape audit is $(1-\rho)^n$, however long the stream. *Proof.* The draw is independent of everything the colluder saw when it acted. □

Planted cases add cheap evidence as long as they cannot be told from real questions, so we make them from real questions by controlled edits, such as changing a timestamp in a port report, with a discriminator that guides the generator until it can no longer tell planted from real (Checco et al., 2018); human audits keep the guarantee when it can, as random tax audits deter evasion when their timing is unpredictable and the evidence survives (Allingham and Sandmo, 1972; Gans and Holden, 2026). Blame is assigned by comparing each agent with the others it was paired with across many random pairings. An agent is flagged when its audited decisions go wrong more often than its track record predicts, or when it does much worse on audited real decisions than on planted ones of matched difficulty, by an anytime-valid sequential test (Ramdas et al., 2023) that keeps the chance that an agent erring no more than predicted is ever flagged below a chosen level α (split across agents), however long monitoring runs; the number of audits needed to flag a colluder then depends on α and on how much more often its decisions go wrong, not on the number of decisions (Steinhardt et al., 2016). Flags lower the agent's weight in Task 1.1 and bar it from arguing or judging; audits never change a decision, and nothing triggers a human audit. Treating auditing as an inspection game, we expect that a colluder that tunes how often it cheats can cause, before it is flagged, wrong decisions beyond the honest error rate that grow at most like the square root of the number of decisions when honest agents also err, and stay bounded when they rarely do; we will prove such bounds and compare them with the red team's best attacks. A colluder that mimics a weak honest agent is never flagged, but a weak record earns little weight and few roles (Task 1.2). We also look for rulings that favor one debater more than the evidence explains (Dong et al., 2009), and paraphrase messages against hidden signals (Roger and Greenblatt, 2023).

Preliminary results. The PI co-advised UT Dallas's team, one of five university red teams in the Amazon Nova AI Challenge (Sahai et al., 2025), which built COMET, a closed-loop red-teaming framework that elicited harmful outputs from guarded code models (Xu et al., 2025). In a pilot, a strong model told templated planted cases from real questions with an AUC of 0.86, but planted cases made by controlled edits only with an AUC of 0.56; in simulation, colluders whose audited decisions went wrong three times as often as their track records predicted were flagged after a median of 40 audited decisions, and no honest agent was flagged ($\alpha = 0.05$).

Evaluation and risks. Against the red team, we measure wrong decisions that get through at each audit budget, time to flag colluders, false flags, and whether a discriminator can tell planted cases from real ones (H4). A group that cheats only once is caught with probability at most ρ, and if planted cases stay recognizable, flags rest on the human audits alone, which slows flagging.

3.4 RG3: Let Agents Self-Improve with Reinforcement Learning on the Group's Decisions

Challenges. Self-improvement needs rewards that track correct answers, and the obvious ones do not: majority agreement rewards conformity and can collapse accuracy (Shafayat et al., 2025), debate wins reward persuasion when the judge can be fooled (Kenton et al., 2026), and approval rewards misleading the evaluator (Wen et al., 2025). Rewards computed from the group's own decisions raise a further worry, that wrong decisions reinforce the behavior that produced them. A group's reward must also say whose messages helped, or training rewards agents that contributed nothing and misses the agent that found the decisive evidence.

Overview. RG3 lowers the agents' own error rates, which can shrink every term of (1). Its rewards need no human labels and inherit the protections of RG1 and RG2: shared mistakes count once, flagged agents are dropped, and judges re-run evidence.

Task 3: Rewards for contributions to trustworthy decisions

Approach. What reward makes agents more accurate when the only labels are the group's own, imperfect decisions? We treat each decision as a noisy label: if it is wrong with probability $\eta_q < \frac{1}{2}$, rewarding agreement with it still pushes an agent toward the right answer, scaled by $1 - 2\eta_q$, provided the reference does not depend on the agent's own answer and agreement by chance earns nothing. We propose rewards built to meet these conditions. *First answers.* Agent a's first answer to q is scored against D_q^{-a}, the answer that the posterior of Task 1.1 favors when a's group is left out, minus the agreement of its answer to another question q' with that reference on a third, q'', both on its topic and drawn at random:

$$r_a(q) = \omega_q \mathbf{1}\{x_{aq} = D_q^{-a}\} - \omega_{q''} \mathbf{1}\{x_{aq'} = D_{q''}^{-a}\}, \tag{5}$$

so an agent earns credit only for agreeing with the rest of the group beyond chance (Dasgupta and Ghosh, 2013; Shnayder et al., 2016), against a reference that counts shared mistakes once and discounts flagged agents.

Proposition 6 (decisions as noisy labels). Consider a yes-or-no question, and suppose that, given the true answer, D_q^{-a} is wrong with probability η_q independently of agent a's answer. Then $\mathbb{E}\mathbf{1}\{x = D_q^{-a}\} = \eta_q + (1 - 2\eta_q)\mathbf{1}\{x = y_q\}$, so in expectation, rewarding agreement with D_q^{-a} gives $1 - 2\eta_q$ times the policy-gradient update that rewarding the true answer would give; the second term of (5) does not depend on x_{aq} and leaves this update unchanged. If a's answers ignore the question, the two terms of (5) have the same expectation over questions, so such answers earn nothing. *Proof.*

$P(D_q^{-a} = x)$ is $1 - \eta_q$ if $x = y_q$ and η_q otherwise, and terms that do not vary with the answer do not change the policy gradient. Answers that ignore the question, x_{aq} and $x_{aq'}$, share one distribution and are independent of (ω_q, D_q^{-a}) and $(\omega_{q''}, D_{q''}^{-a})$, which also share one distribution. □

So a wrong decision shrinks the update but does not reverse it while $\eta_q < \frac{1}{2}$, and $\omega_q = 1 - 2\hat{\eta}_q$ (with more answers, $1 - \hat{\eta}_q|\mathscr{Y}|/(|\mathscr{Y}| - 1)$), where $\hat{\eta}_q$ comes from the left-out posterior, weights each question by that push: confident references count fully, near-ties little. Independence fails when the reference includes the agent's own answer, since the agent can then raise its reward by agreeing with itself and agents like it, which is how rewards for majority agreement collapse (Shafayat et al., 2025); leaving out the agent's group restores it under the model of Task 1.1, and subtracting chance agreement removes the reward for answers that ignore the question, such as always saying yes, even when colluders coordinate on them. For score bands, exact agreement gives way to the correlated-agreement score (Shnayder et al., 2016).

Debate messages. Each message m_t is credited by how far it moves the judge toward the decision:

$$r_t = \omega_q \left[p_J(D_q \mid m_{1:t}) - p_J(D_q \mid m_{1:t-1}) \right], \tag{6}$$

where p_J is the judge's probability of the decision D_q given the debate so far, obtained by re-scoring each prefix of the logged transcript, and $\hat{\eta}_q$ here comes from that probability, calibrated on development questions. The credits of a debate sum to ω_q times the change in the judge's probability of the decision, so (6) splits that change among messages, telescoping as potential-based shaping does (Ng et al., 1999) (its potential depends on the decision, so policy invariance does not carry over), and credits each message with its effect given the earlier ones (Wolpert and Tumer, 2001; Agogino and Tumer, 2008). When the judge decides by re-running evidence, the decision does not depend on persuasion, so messages earn credit for moving the judge toward what the evidence shows. At scale, we will train a multi-agent process reward model on logged debates to predict each message's credit, with labels such as "introduced re-runnable evidence" and "persuaded without evidence," extending our FG-PRM, and compare it with order-free credit such as Shapley values. Decisions involving flagged agents are dropped, planted cases never enter training, agents are rewarded separately, agents that judge each other never share a reward, no judge is trained on its own rulings (Shi et al., 2026), and training uses policy-gradient methods (Shao et al., 2024).

Preliminary results. Our FG-PRM labels each step of a math solution with one of six hallucination types and, as a verifier, substantially improves accuracy on GSM8K and MATH (Li et al., 2025); our HiPRAG rewards each search decision of an agent, not only its final answer (Wu et al., 2026). In a pilot training a 1.5B-parameter model on GSM8K with GRPO (Shao et al., 2024), with three of eight reference agents sharing a flawed hint, rewarding majority agreement raised held-out accuracy from 0.62 to 0.68 and then collapsed it to 0.41 within 300 steps, while reward (5) raised it to 0.73 without collapse: 73% of the gain from training on true answers (to 0.77), close to the $1 - 2\hat{\eta} = 0.74$ that Proposition 6 predicts.

Evaluation and risks. On all three testbeds, we compare our rewards with rewards for majority agreement and for debate wins (Arnesen et al., 2024) at matched compute, on open-weight models from several families, measuring the accuracy of decisions and first answers on held-out questions, topics, and model families, including where human judges are unreliable (Burns et al., 2024), and dishonest behavior during training: invented evidence, unsupported changes of answer, and wrong decisions the red team gets through (H5). A misconception most agents share makes η_q exceed one

Table 3: Testbeds. Each provides known answers, controlled shared mistakes, and tools whose results the judge can re-run.

Testbed	Questions and sources	Shared mistakes and dishonesty	Tools in the sandbox
A. Math and code	GSM8K and MATH problems (Cobbe et al., 2021; Hendrycks et al., 2021); programming problems with reference outputs	Some agents receive the same flawed hint, so the groups that share mistakes are known; dishonest agents	Calculator, code execution
B. Multi-document questions	Questions composed from event structures by our MEQA pipeline (Li et al., 2024b), including synthetic maritime scenarios whose reports form source trees of originals, copies, paraphrases, and outdated versions	Circular reporting built into the source trees; forged reports and spoofed position tracks; a compromised agent	Document search; a port and vessel database built from the same events
C. Scientific claims and peer review	Whether a paper's evidence supports its claims, on public papers with their OpenReview reviews and rebuttals, and the PeerRead corpus (Kang et al., 2018)	Planted errors in papers (Schroter et al., 2008); AI reviewers that share an underlying model	Paper search; code execution for released code

half on some questions, and peer prediction can pay for it (Denisov-Blanch et al., 2026); accuracy on held-out questions with known answers detects this, and training stops if it falls. Training on the same decisions can make agents alike, so we measure shared error factors before and after training and train agents on different subsets of decisions. Debaters could learn to join the judge's likely decision instead of defending their answer; if so, we will credit messages for moving the judge toward the debater's own answer, which makes (6) a shaped debate-win reward.

3.5 Evaluation, Expected Results, and Limitations

All data are public or synthetic, and every question has a known answer for scoring, kept apart from the planted cases agents may meet. In experiments, a human audit reveals the known answer of a decision drawn at random; we also study audits that are sometimes wrong (Table 3).

A peer-review example. In testbed C, two AI reviewers disagree on whether a paper's experiments support its claimed five-point gain; the doubter disputes the step "the baseline was evaluated on the same data split," and the judge re-runs the released evaluation script for both settings, so an area chair sees the disputed step and the reproduced numbers.

Agent pools and red team. Agents are built on open-weight models of several families and sizes, plus commercial models where licenses allow, with varied role instructions and tools. In testbed C, published reproducibility reports supply known answers for human audits (Pineau et al., 2021), and agents are not told which claims have reports, so Proposition 5 holds within that set. Our red team, modeled on AI control evaluations (Greenblatt et al., 2024), builds attackers of increasing strength: (1) a fixed dishonest agent; (2) an agent that knows the protocol but not its random draws;

Table 4: Schedule by half-year (Y1 = April 2027 – March 2028) and milestones.

Activity	Y1a	Y1b	Y2a	Y2b	Y3a	Y3b
Testbeds A and B; red-team levels 1–2	■	■				
Testbed C; red-team level 3			■	■		
Task 1.1: shared-factor model, identifiability; then topics, stakes	■	■	■			
Task 1.2: impartial selection; then coalitions, price of impartiality		■	■	■		
Task 2.1: harness v1; then false-step search, right of reply	■	■	■	■		
Task 2.2: planted and human audits; then audit bounds			■	■	■	
Task 3: rewards (5)–(6); then credit model, comparison				■	■	■
Red-team level 4; full evaluation; release					■	■

Milestones. M1 (month 6): harness v1 and testbed A released. M2 (12): shared-factor model and its identifiability analysis, on testbeds A and B. M3 (18): impartial selection integrated; tool-backed debate on testbed B. M4 (24): audit scheme and bounds against red-team levels 1–3; testbed C. M5 (30): analysis and comparison of rewards. M6 (36): full framework against level 4; code, testbeds, and protocols released.

(3) an agent trained to spot planted debates that cheats rarely, and only when it expects no audit; and (4) a debater and a judge that coordinate through hidden signals. We report wrong decisions that get through at each audit budget, together with how much useful work the framework still does, as safety–usefulness curves.

Conditions varied. Each hypothesis predicts where its method should help and where it should fail, so experiments vary the conditions that decide this: (i) how widely mistakes are shared, from no agents to most given the same flawed hint or outdated report, and the number of model families, from one to five; (ii) dishonesty, from none to 30% of agents, at red-team levels 1–4; (iii) evidence, the share of disputed steps a tool can settle and of reports that are outdated or forged; (iv) budgets for debates, audits, and computation, including small models and few tool calls, as under limited connectivity; and (v) answer types, from yes or no to score bands. Regimes where the propositions say the methods fail are included, so failures are measured, not assumed; transfer is measured on held-out topics and model families.

Metrics, baselines, and statistics. Primary metrics are decision accuracy and calibration, wrong decisions under attack, correct minorities recovered and correct majorities overturned, audits needed to flag colluders, and cost in computation and tool calls. Each term of (1) is measured separately, so a gain can be traced to the task that produced it. Baselines are a single AI judge, majority vote, and averaging with equal weights, at matched compute; ablations remove each component in turn. We report means with confidence intervals over seeds and question samples.

Results and new knowledge. If successful, the project will produce (i) a rater model that counts a group sharing a mistake at most once, with conditions for learning it without an answer key and rates for detecting groups; (ii) an impartial mechanism for choosing debaters and judges, with its measured price and bounds on coalition gains; (iii) an open agentic debate harness that reduces judging a question to judging single steps, with the judge's error measured for each kind of step; (iv) an audit scheme with bounds on the harm a colluder can do before it is flagged; and (v) a self-improvement method whose expected update follows accuracy rather than agreement. Scientifically, it will show when a debate among AI agents can be trusted and how few audits keep the agents honest; because people audit only a few decisions and training needs no labels, the methods are

Table 5: Goals, benchmarks (testbeds A–C, PeerEval), and success criteria, fixed before the experiments.

Goal and benchmark	Success criterion
G1 (Task 1.1). Accuracy, calibration, and recovery of seeded groups across levels of sharing (A, B)	A margin over majority vote that grows with how widely mistakes are shared, where three or more groups err independently (H1, Year 2)
G2 (Task 1.2). Selection gain of strategic raters and coalitions; price of impartiality (PeerEval, C)	No gain for a single rater; coalition gains below a preset bound; impartiality costs less than debate adds (H2, Year 2)
G3 (Task 2.1). Accuracy on disputed questions; judge's error on each kind of step (B, C)	More accurate than open discussion and all baselines where a tool can settle the disputed steps (H3, Year 2)
G4 (Task 2.2). Wrong decisions that get through, time to flag, false flags, at red-team levels 1–4	Colluders cheating at a fixed rate flagged after a number of audits that does not grow with the stream; false flags below the chosen level (H4, Year 3)
G5 (Task 3). Held-out accuracy of decisions and first answers; rates of dishonest behavior	Larger gains than agreement and debate-win rewards, with no rise in dishonesty (H5, Year 3)
G6. Code, testbeds, protocols; peer-reviewed papers with proofs	Public release by month 36

built to scale to AI whose answers people cannot easily evaluate.

Limitations. Where no tool applies, the judge rules on argument alone. A misconception every agent shares cannot be detected from answers alone (Proposition 2); a group that cheats once is caught at most with the audit probability (Proposition 5); paraphrasing limits but does not remove hidden signals (Mathew et al., 2025); and Navy use would need Navy data and expert validation.

3.6 Project Schedule, Milestones, Goals, and Benchmarks

The project runs 36 months from April 1, 2027; Table 4 gives the schedule by half-year and milestones M1–M6.

Goals and benchmarks. Each goal has a benchmark and a criterion for success (Table 5); results that miss the criteria will be reported as findings.

3.7 Gold Standard Science

The project will follow the tenets of Executive Order 14303, as the PI's previous work does.

Reproducible and transparent. The harness logs every debate with its exact tool calls, model versions, prompts, and seeds, and tool evidence is re-runnable by design. Code, testbeds, prompts, and protocols will be released with each paper, and the PI's group studies reproducibility (Yan et al., 2025).

Communicative of error and uncertainty. Every guarantee is stated with its assumptions, decisions carry calibrated probabilities, and the framework's accuracy is estimated from random human audits with valid confidence intervals (Angelopoulos et al., 2023).

Skeptical and falsifiable. A red team attacks our own methods, ablations remove each component, and each task states a hypothesis that experiments can refute (H1–H5, Table 2) and the regime in

which its method should fail. Metrics and analysis plans will be fixed before the main experiments.

Collaborative, peer reviewed, and open to negative results. The work joins statistics, social choice, AI safety, and natural language processing, with results for peer-reviewed venues in each; failures of debate or audits will be published as findings.

Without conflicts of interest. The PI will disclose outside activities and support as UT Dallas policy requires, and evaluations use public or synthetic data with released code.

4 Management Approach

This is a single-PI award with no subawardees, co-investigators, or senior personnel. The PI will direct all research and be responsible for schedules, reports, and releases. The budget supports two Ph.D. students for three years, one month of PI summer effort per year, travel to ONR Project and Program Review Meetings and to conferences, and computing.

Team and roles. Student 1 leads the statistical and mechanism-design work and its proofs: Tasks 1.1 and 1.2, the audit analysis of Task 2.2, and testbeds A and C. Student 2 leads the agent and learning work: the harness of Task 2.1, Task 3, and testbed B, which reuses the MEQA pipeline. Both students run the red team, rotating roles so that no one attacks only their own method. Code, logs, and planted-case generators live in one shared repository, with the harness frozen at each milestone so that results stay comparable.

Reporting and dissemination. The PI will submit annual and final reports, attend Project and Program Review Meetings, publish in peer-reviewed venues, and release code, testbeds, and protocols at M1, M6, and with each paper.

Relation to other support. The PI's NSF CAREER award studies event knowledge from long documents, the subject of the MEQA pipeline that testbed B reuses; the Open Philanthropy grant supports control evaluations of AI debate; the Amazon Research Award concerns process-guided fine-tuning. This proposal funds aims those awards do not, and any overlap with a pending proposal would be removed with the program officers.

5 Principal Investigator Qualifications

Position and training. The PI, Xinya Du, has been an Assistant Professor of Computer Science at The University of Texas at Dallas since 2022, in a first tenure-track appointment. The PI received a Ph.D. in Computer Science from Cornell University in 2021, advised by Claire Cardie, and was a postdoctoral researcher with Heng Ji at the University of Illinois Urbana-Champaign, where the PI was first author of RESIN-11, a schema-guided event prediction system in DARPA's KAIROS program (Du et al., 2022). The PI was named a University of Chicago Rising Star in Data Science in 2021.

Recognition and service. NSF CAREER Award (2024–2029); Amazon Research Award (2023); Cisco Research Award (2025). The PI's Ph.D. paper on neural question generation (Du et al., 2017) has been cited more than 700 times and is ranked by Paper Digest among the most influential ACL 2017 papers. The PI has served as an area chair for ACL (2023, 2024, 2025), EMNLP (2024, 2025), LREC-COLING 2024, and COLING 2025, and as Website Chair of NAACL 2024, and co-organized the AI4Research workshop at AAAI 2025 and the Workshop on Multimodal Knowledge and Language Modeling at IJCAI 2025.

Independent contributions since the Ph.D. With students at UT Dallas, the PI has built a line of work on AI evaluating AI that this project extends.

- *Peer evaluation among AI judges.* PRD (Li et al., 2024a) introduced peer ranking and peer discussion among AI judges, which agree with human judgments better than the strongest single judge and reduce a judge's bias toward its own model. It underlies PeerEval (peereval-ai.github.io), a public leaderboard on which seven AI models answer the same questions and judge one another anonymously (40,156 pairwise judgments in its first release). IQA-EVAL (Li et al., 2024c) evaluates interactive question answering with persona-driven AI agents.
- *Rewards and verification.* FG-PRM (Li et al., 2025) detects six types of hallucination at each step of a math solution; HiPRAG (Wu et al., 2026) rewards each search decision of an agent, making agentic search more accurate and efficient; and Liang et al. (2026) show that a reasoning model can verify answers without further training.
- *Benchmarks and AI for science.* MEQA (Li et al., 2024b) provides event-centric multi-hop questions with explanations, FaithScore (Jing et al., 2024) measures fine-grained hallucinations, and LMR-Bench (Yan et al., 2025) evaluates agents that reproduce research code; the PI co-wrote a survey of LLMs for scientific research, including peer review (Luo et al., 2025), and the group's research-idea generator appears at EMNLP 2026 (Li et al., 2026a).
- *Adversarial evaluation.* The PI co-advised UT Dallas's team, one of the five university red teams in the Amazon Nova AI Challenge (Sahai et al., 2025), which built the COMET red-teaming framework (Xu et al., 2025), and holds an Open Philanthropy grant on control evaluations of AI debate.

This work has appeared at NeurIPS, ICLR, ACL, EMNLP, and TMLR.

6 Responsibility

Resources. UT Dallas provides the PI's laboratory space, a GPU server (six A100 and four RTX A6000 GPUs), and the shared Juno cluster (67 GPUs; see Facilities and Other Resources). With the requested cloud and model-access costs, these suffice; no equipment purchase is requested.

Ability to comply. The UT Dallas Office of Research and Innovation administers the PI's federal awards and will administer this award under 2 CFR 200 and ONR terms; the PI's commitments, listed in Current and Pending Support, leave the proposed effort available.

Performance history. The PI has managed an NSF CAREER award (2024–2029) and industry and foundation awards, filing reports on time; funds on current awards are being spent on schedule.

Integrity and business ethics. The PI and UT Dallas follow federal research-integrity policies (DoDI 3210.7); the PI has no record of misconduct.

Qualifications and eligibility. The PI is a U.S. permanent resident, has held a full-time tenure-track appointment at UT Dallas since 2022, the PI's first, and received the Ph.D. from Cornell University in 2021, on or after January 1, 2019, as Section B of the NOFO requires.

Organization and controls. UT Dallas maintains federally reviewed accounting, property-control, research-security, quality-assurance, and safety systems and is registered in SAM (UEI EJCVPNN1WFS5; CAGE 0W921); the research is computational.

Xinya Du

Contact

Assistant Professor
Department of Computer Science
The University of Texas at Dallas
800 West Campbell Road, Richardson, TX 75080

Email: xinya.du@utdallas.edu
Website: https://xinyadu.github.io
Google Scholar
Phone: 607-319-6425

Research Interests

Natural Language Processing, Large Language Models, AI for Science.

Education

Cornell University — Aug 2016 – Aug 2021
Ph.D. in Computer Science (M.S. degree granted in Aug 2019)
Advisor: Claire Cardie. Worked closely with Sasha Rush.

Shanghai Jiao Tong University — Sep 2012 – Aug 2016
B.E. in Computer Science and Engineering
Outstanding Graduates Award

Professional Experience

University of Texas at Dallas — Aug 2022 – Present
Assistant Professor in Computer Science

University of Illinois at Urbana-Champaign — Sep 2021 – Aug 2022
Postdoctoral Research Associate. Worked with Heng Ji.

Industrial Experience

Google AI Research — May 2020 – Aug 2020
Research Intern. Supervisor: Luheng He, Qi Li, Emily Pitler.

Allen Institute for Artificial Intelligence (AI2) — Sep 2018 – Dec 2018
Research Intern. Supervisor: Bhavana Dalvi, Peter Clark.

Microsoft Research (MSR) — May 2018 – Aug 2018
Research Intern. Supervisor: Paul Bennett, Ahmed Awadallah.

Awards & Honors

Faculty Research Award — UT Dallas CS, 2025
NSF CAREER award — National Science Foundation, 2024
Best Paper Award — AAAI AI4Research Workshop, 2025
Best Poster Award (Top 1%) — ICML AI for Science, 2024
AAAI New Faculty Highlights — AAAI, 2024
Cisco Faculty Research Award — Cisco, 2024
President's Teaching Excellence Awards Nominee — UT Dallas, 2024
Amazon Research Award — Amazon, 2023
Spotlight Rising Star in Data Science — University of Chicago, 2021
Top 100 New Stars in Artificial Intelligence — Baidu Scholar, 2020
Most Influential ACL Papers (15 each year) — Paperdigest, 2017

National Scholarship (Top 1% students nationwide) SJTU, 2013

National Olympiad in Mathematics in Provinces, First Prize 2011

National Olympiad in Informatics in Provinces, First Prize 2010

Grants

Control Evaluations for AI Debate (Single PI)
Funding Source: Open Philanthropy
Period: Oct 2025 – present.
Amount Awarded: $218,513 (+ $40,000 Computing). Amount to me: $218,513.

NSF CAREER: Learning to Extract Consistent Event Graphs from Long and Complex Documents (Single PI)
Funding Source: National Science Foundation (NSF).
Period: May 2024 – present.
Amount Awarded: $561,219. Amount to me: $561,219.

Adobe Research Gift: Hallucinations in Video Language Models (Single PI)
Funding Source: Adobe.
Period: Sep 2025 – present.
Amount Awarded: $30,000.

Process-guided Fine-tuning for Answering Complex Questions (Single PI)
Funding Source: Amazon Research Award.
Period: Jan 2024 – present.
Amount Awarded: $110,000 ($40,000 AWS credits). Amount to me: $110,000.

Amazon Trusted AI Challenge Grant (Co-PI, with Prof. Wei Yang)
Funding Source: Amazon.
Period: Jan 2025 – present.
Amount Awarded to Our Team: $250,000.

Optimizing RISC-V Compilers with RISC-LLM and Syntax Parsing (Lead Contributor; PI: Wei Yang)
Funding Source: Amazon Research Award.
Period: Sep 2025 – Aug 2026.
Amount Awarded: $70,000. Amount to me: $23,000.

FAIGen: Faithful LLM Generation with Scientific Principles-guided Learning (Single PI)
Funding Source: Cisco Faculty Research Award.
Period: Jan 2025 – present.
Amount Awarded: $90,000. Amount to me: $90,000.

Mitigating Hallucinations in Vision-language Models with Fine-grained AI Feedback (Single PI)
Funding Source: OpenAI Researcher Access Program.
Period: Sep 2024 – present.
Amount Awarded: $12,000. Amount to me: $12,000.

Undergraduate Research Apprenticeship Award (URAP) (Single PI)
Funding Source: University of Texas at Dallas.
Period: Summer 2023, 2024, 2025.
Amount Awarded: $13,500. Amount to me: $13,500.

New Faculty Research Symposium Grant (Leading PI)
Funding Source: University of Texas at Dallas.
Period: Aug 2025 – Jul 2026.
Amount Awarded: $25,000.

Grants (Pending)

Collaborative Research: RI: SMALL: Explainable Methods for Natural Language Reasoning (Leading PI, co-PI: Prof. Claire Cardie)
Funding Source: National Science Foundation (NSF).
Period: Jan 2025 – Dec 2028.
Amount Requested: $600,000. Amount to me: $300,000.
Status: Pending (**rated highly competitive on first submission**).

ACED: Large Language Model-based Research Agent for Synthetic Biology Research Idea Generation (Leading PI, co-PI: Prof. Leo Bleris)
Funding Source: National Science Foundation (NSF).
Period: Aug 2026 – Dec 2028.
Amount Requested: $600,000. Amount to me: $300,000.
Status: Pending (**rated competitive on first submission**).

III: Automating Scientific Knowledge Discovery through Retrieval, Generation, and Validation (Leading PI, co-PI: Prof. Vincent Ng)
Funding Source: National Science Foundation (NSF).
Period: Jan 2027 – Dec 2029.
Amount Requested: $600,000. Amount to me: $300,000.
Status: Pending.

LLM-Guided Agents for Rational and Reliable Cell Reprogramming (Co-PI, co-PI: Prof. Leo Bleris)
Funding Source: National Science Foundation (NSF) Direct For Biological Sciences (BIO).
Period: Aug 2026– Dec 2029.
Amount Requested: $726,459. Amount to me: $242,153.
Status: Pending.

Publications

Note: † indicates that I am a co-leading author. * indicates equal contributions.
You can also find my publication list on [DBLP] and [Google Scholar] pages.

[1] **LDC: Learning to Generate Research Idea with Dynamic Control** [pdf]
Ruochen Li, Liqiang Jing, Chi Han, Jiawei Zhou, **Xinya Du**
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026. Earlier version in AAAI 2025 AI4Research Workshop. **(Oral)**
Best Paper Award at AAAI AI4Research.

[2] **A Comprehensive Analysis for Visual Object Hallucination in Large Vision-Language Models** [pdf]
Liqiang Jing, Hardy Chen, Ehsan Aghazadeh, Xin Eric Wang, **Xinya Du**
In Findings of the Association for Computational Linguistics: (EMNLP), 2026.

[3] **A Skill-augmented Agentic Framework and Benchmark for Multi-Video Understanding** [pdf]
Yue Zhang, Liqiang Jing, Jia Li, Yapeng Tian, **Xinya Du**, Yunhui Guo, Vibhav Giridhar Gogate
In Findings of the Association for Computational Linguistics: (EMNLP), 2026.

[4] **AMARIS: A Memory-Augmented Rubric Improvement System for Rubric-Based Reinforcement Learning** [pdf]
Peilin Wu, Xinlu Zhang, Kun Wan, Wentian Zhao, Gang Wu, **Xinya Du**, Zhiyu Chen
In Findings of the Association for Computational Linguistics: (EMNLP), 2026.

[5] **HiPRAG: Hierarchical Process Rewards for Efficient Agentic Retrieval Augmented Generation** [pdf]
Peilin Wu, Mian Zhang, Kun Wan, Wentian Zhao, Kaiyu He, **Xinya Du**, Zhiyu Chen
In International Conference on Learning Representations (ICLR), 2026.

[6] **Is Grokking Worthwhile? Functional Analysis and Transferability of Generalization Circuits in Transformers** [pdf]
Kaiyu He, Mian Zhang, Peilin Wu, **Xinya Du**, Zhiyu Chen
In Findings of the Association for Computational Linguistics: (ACL), 2026.

[7] **Your Reasoning Model is Secretly a Reward Model – Optimization-Free Verification from Experience** [pdf]
Zhenwen Liang, Ruosen Li, Yujun Zhou, Linfeng Song, Dian Yu, **Xinya Du**, Haitao Mi, Dong Yu
In Annual Meeting of the Association for Computational Linguistics (ACL), 2026.

[8] **FIFA: Unified Faithfulness Evaluation Framework for Text-to-Video and Video-to-Text Generation** [pdf]
Liqiang Jing, Viet Dac Lai, Seunghyun Yoon, Trung Bui, **Xinya Du**
In Findings of the Association for Computational Linguistics: (ACL), 2026.

[9] **Dr.V: A Hierarchical Perception-Temporal-Cognition Framework to Diagnose Video Hallucination by Fine-Grained Spatial-Temporal Grounding** [pdf]
Meng Luo, Shengqiong Wu, Liqiang Jing, Tianjie Ju, Li Zheng, Jinxiang Lai, Tianlong Wu, **Xinya Du**, Jian Li, Siyuan Yan, Jiebo Luo, William Yang Wang, Hao Fei, Mong-Li Lee, Wynne Hsu
In International Journal of Computer Vision (IJCV), 2026.

[10] **LMR-Bench: Evaluating LLM Agent's Ability on Reproducing Language Modeling Research** [pdf]
Shuo Yan, Ziming Luo, Zimu Wang, Ruochen Li, Daoyang Li, Liqiang Jing, Kaiyu He, Peilin Wu, Juntong Ni, George Michalopoulos, Yue Zhang, Ziyang Zhang, Mian Zhang, Zhiyu Chen, **Xinya Du**
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025.

[11] **FG-PRM: Fine-grained Hallucination Detection and Mitigation in Language Model Mathematical Reasoning** [pdf]
Ruosen Li, Ziming Luo, **Xinya Du**
In Findings of the Association for Computational Linguistics: (EMNLP), 2025.

[12] **SciEvent: Benchmarking Multi-domain Scientific Event Extraction** [pdf]
Bofu Dong, Pritesh Shah, Sumedh Sonawane, Tiyasha Banerjee, Erin Brady, **Xinya Du**, Ming Jiang
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025.

[13] **Search Wisely: Mitigating Sub-optimal Agentic Searches By Reducing Uncertainty** [pdf]
Peilin Wu, Mian Zhang, Xinlu Zhang, **Xinya Du**, Zhiyu Chen
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025.

[14] **SFT or RL? An Early Investigation into Training R1-Like Reasoning Large Vision-Language Models** [pdf]
Hardy Chen, Haoqin Tu, Fali Wang, Hui Liu, Xianfeng Tang, **Xinya Du**, Yuyin Zhou, Cihang Xie
In Transactions on Machine Learning Research (TMLR), 2025.

[15] **FGAIF: Aligning Large Vision-Language Models with Fine-grained AI Feedback** [pdf]
Liqiang Jing, **Xinya Du**
In Transactions on Machine Learning Research (TMLR), 2025.

[16] **VistaDPO: Video Hierarchical Spatial-Temporal Direct Preference Optimization for Large Video Models** [pdf]
Haojian Huang*, Haodong Chen*, Shengqiong Wu, Meng Luo, Jinlan Fu,
Xinya Du, Hanwang Zhang, Hao Fei
In International Conference on Machine Learning (ICML), 2025.

[17] **DSBench: How Far Are Data Science Agents to Become Data Science Experts?** [pdf]
Liqiang Jing, Zhehui Huang, Xiaoyang Wang, Wenlin Yao, Wenhao Yu, Kaixin Ma, Hongming Zhang, **Xinya Du**, Dong Yu
In International Conference on Learning Representations (ICLR), 2025.

[18] **FIHA: Autonomous Hallucination Evaluation in Vision-Language Models with Davidson Scene Graphs** [pdf]
Bowen Yan, Zhengsong Zhang, Liqiang Jing, Eftekhar Hossain and **Xinya Du**
In Findings of the Association for Computational Linguistics: (ACL), 2025.

[19] **Efficient Document-level Event Relation Extraction**
Ruochen Li, Zimu Wang, **Xinya Du**
In 10th Workshop on Representation Learning for NLP at Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2025.

[20] **PRD: Peer Rank and Discussion Improve Large Language Model-based Evaluations** [pdf]
Ruosen Li, Teerth Patel, **Xinya Du**
In Transactions on Machine Learning Research (TMLR), 2024.

[21] **Large Language Models for Automated Open-domain Scientific Hypotheses Discovery** [pdf]
Zonglin Yang, **Xinya Du**†, Junxian Li, Jie Zheng, Soujanya Poria, Erik Cambria
In Findings of the Association for Computational Linguistics: (ACL), 2024.
Best poster award in ICML AI4Science, 2024 (Top 1% of 250 submissions).

[22] **Multi-hop Event-centric Question Answering with Explanations**
Ruosen Li, Zimu Wang, Son Quoc Tran, Lei Xia, **Xinya Du**
In Annual Conference on Neural Information Processing Systems (NeurIPS), 2024.

[23] **FaithScore: Fine-grained Evaluations of Hallucinations in Large Vision-Language Models** [pdf]
Liqiang Jing, Ruosen Li, Yunmo Chen, **Xinya Du**
In Findings of the Association for Computational Linguistics: (EMNLP), 2024.

[24] **IQA-EVAL: Automatic Evaluation of Human-Model Interactive Question Answering** [pdf]
Ruosen Li, Ruochen Li, Barry Wang, **Xinya Du**
In Annual Conference on Neural Information Processing Systems (NeurIPS), 2024.

[25] **QAEvent: Event Extraction as Question-Answer Pairs Generation** [pdf]
Milind Choudhary, **Xinya Du**
In Findings of the Association for Computational Linguistics: (EACL), 2024.

[26] **Language Models as Inductive Reasoners** [pdf]
Zonglin Yang, Li Dong, **Xinya Du**†, Hao Cheng, Erik Cambria, Xiaodong Liu, Jianfeng Gao, Furu Wei
In Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2024. **(Oral)**

[27] **Making Natural Language Reasoning Explainable and Faithful** [pdf]
Xinya Du
In AAAI Conference on Artificial Intelligence (AAAI), 2024. **(Oral)**

[28] **Proto-CLIP: Vision-Language Prototypical Network for Few-Shot Learning** [pdf]
Jishnu Jaykumar P, Kamalesh Palanisamy, Yu-Wei Chao, **Xinya Du**, Yu Xiang
In International Conference on Intelligent Robots and Systems (IROS), 2024. **(Oral)**

[29] **Document-level Causal Relation Extraction with Knowledge-guided Binary Question Answering** [pdf]
Zimu Wang, Lei Xia, Wei Wang, **Xinya Du**
In Findings of the Association for Computational Linguistics: (EMNLP), 2024.

[30] **Leveraging Structured Information for Explainable Multi-hop Question Answering and Reasoning** [pdf]
Ruosen Li, **Xinya Du**
In Findings of the Association for Computational Linguistics: (EMNLP), 2023.

[31] **Process of Elimination for Multiple Choice Reasoning** [pdf]
Chenkai Ma, **Xinya Du**
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2023.

[32] **Probing Representations for Document-level Event Extraction** [pdf]
Barry Wang, **Xinya Du**, Claire Cardie
In Findings of the Association for Computational Linguistics: (EMNLP), 2023.

[33] **End-to-end Case-Based Reasoning for Commonsense Knowledge Base Completion** [pdf]

Zonglin Yang, **Xinya Du**†, Erik Cambria, Claire Cardie
In Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2023. **(Oral)**

[34] **Zero-Shot Classification by Logical Reasoning on Natural Language Explanations** [pdf]
Chi Han, Hengzhi Pei, **Xinya Du**, Heng Ji
In Findings of the Association for Computational Linguistics: (ACL), 2023.

[35] **Toward Consistent and Informative Event-Event Temporal Relation Extraction** [pdf]
Xiaomeng Jin, Haoyang Wen, **Xinya Du**, Heng Ji *In MATCHING at Annual Meeting of the Association for Computational Linguistics (ACL), 2023.* **(Oral)**

[36] **Logical Entity Representation in Knowledge-Graphs for Differentiable Rule Learning** [pdf]
Chi Han, Qizheng He, Charles Yu, **Xinya Du**, Hanghang Tong, Heng Ji
In International Conference on Learning Representations (ICLR), 2023.

[37] **RESIN-11: Schema-guided Event Prediction for 11 Newsworthy Scenarios** [pdf]
Xinya Du, Zixuan Zhang, Sha Li, Heng Ji, the RESIN team
In Conference of the North American Chapter of the Association for Computational Linguistics (NAACL): System Demonstrations, 2022.
Top ranking system in DARPA KAIROS evaluation. [Link]

[38] **Retrieval-Augmented Generative Question Answering for Event Argument Extraction** [pdf]
Xinya Du, Heng Ji
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2022.

[39] **Dynamic Global Memory for Document-level Argument Extraction** [pdf]
Xinya Du, Sha Li, Heng Ji
In Annual Meeting of the Association for Computational Linguistics (ACL), 2022.

[40] **Automatic Error Analysis for Document-level Information Extraction** [pdf]
Aliva Das*, **Xinya Du***, Barry Wang*, Kejian Shi, Jiayuan Gu, Thomas Porter, Claire Cardie
In Annual Meeting of the Association for Computational Linguistics (ACL), 2022.

[41] **Template Filling with Generative Transformers** [pdf]
Xinya Du, Alexander M. Rush, Claire Cardie
In Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2021.

[42] **GRIT: Generative Role-filler Transformers for Document-level Event Entity Extraction** [pdf] **Xinya Du**, Alexander M. Rush, Claire Cardie
In Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2021. **(Oral)**

[43] **Few-shot Intent Classification and Slot Filling with Retrieved Examples** [pdf]
Dian Yu, Luheng He, Yuan Zhang, **Xinya Du**, Panupong Pasupat, Qi Li *In Conference of the North American Chapter of the Association for Computational Linguistics (NAACL),*

2021. **(Oral)**

[44] **QA-Driven Zero-shot Slot Filling with Weak Supervision Pretraining** [pdf]
Xinya Du, Luheng He, Qi Li, Dian Yu, Panupong Pasupat, Yuan Zhang
In Annual Meeting of the Association for Computational Linguistics (ACL), 2021.

[45] **Event Extraction by Answering (Almost) Natural Questions** [pdf]
Xinya Du, Claire Cardie
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2020. **(Oral)**
Top 1% most cited articles published in Computer Science in 2020.

[46] **Improving Event Duration Prediction via Time-aware Pre-training** [pdf]
Zonglin Yang, **Xinya Du**, Alexander M. Rush, Claire Cardie
In Findings of the Association for Computational Linguistics: (EMNLP), 2020.

[47] **Document-Level Event Role Filler Extraction using Multi-Granularity Contextualized Encoding** [pdf]
Xinya Du, Claire Cardie
In Annual Meeting of the Association for Computational Linguistics (ACL), 2020.

[48] **Leveraging Structured Metadata for Improving Question Answering on the Web** [pdf]
Xinya Du, Adam Fourney, Robert Sim, Claire Cardie, Paul Bennett, Ahmed Hassan Awadallah
In Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (AACL/IJCNLP), 2020.

[49] **Be Consistent! Improving Procedural Text Comprehension using Label Consistency** [pdf]
Xinya Du, Bhavana Dalvi, Niket Tandon, Antoine Bosselut, Wen-tau Yih, Peter Clark, Claire Cardie
In Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2019.

[50] **Harvesting Paragraph-Level Question-Answer Pairs from Wikipedia** [pdf]
Xinya Du, Claire Cardie
In Annual Meeting of the Association for Computational Linguistics (ACL), 2018. **(Oral)**
Top 1% most cited articles published in Computer Science in 2018.

[51] **Identifying Where to Focus in Reading Comprehension for Neural Question Generation** [pdf]
Xinya Du, Claire Cardie
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2017.

[52] **Learning to Ask: Neural Question Generation for Reading Comprehension** [pdf]
Xinya Du, Junru Shao, Claire Cardie
In Annual Meeting of the Association for Computational Linguistics (ACL), 2017.
Most Influential ACL Papers (15 each year). [Link]
Featured in New Scientist and TechRepublic. [Link]
Top 0.1% most cited articles published in Computer Science in 2017.

[53] **Cornell Belief and Sentiment System at TAC 2016** [pdf]
Vlad Niculae, Kai Sun, Xilun Chen, Yao Cheng, **Xinya Du**, Esin Durmus, Arzoo Katiyar, Claire Cardie
In Text Analysis Conference (TAC), 2016.

Surveys and Challenge Reports

[54] **LLM4SR: A Survey on Large Language Models for Scientific Research** [pdf]
Ziming Luo, Zonglin Yang, Z. Xu, Wei Yang, **Xinya Du**
arXiv:2501.04306, 2025.

[55] **COMET: Closed-loop Orchestration for Malicious Elicitation Techniques in Code Models** [pdf]
Z. Xu, T. Li, R. S. Rathnasuriya, Z. Song, J. Ren, B. Mandalapu, S. Setayeshpour, **Xinya Du**, Wei Yang
In 1st Proceedings of Amazon Nova AI Challenge (Trusted AI), 2025.

Mentoring Experience

PhD students

Ruosen Li (2022–2026, UT Dallas PhD student, graduated Aug 2026).
Publications: TMLR 2024, NeurIPS 2024, EMNLP Findings 2023 2024.

Liqiang Jing (2023–present, UT Dallas PhD student).
Publications: EMNLP Findings 2024, TMLR 2025, ICLR 2025, EMNLP 2025.
Award: OpenAI Researcher Access Program.

Ruochen Li (2023–present, UT Dallas PhD student).
Publications: NeurIPS 2024, EMNLP 2025.

Master Students

Milind Choudhary (2023, UT Dallas Master student) → UT Dallas PhD.
Topic: Event Extraction as Question-Answer Pairs Generation.
Publications: EACL 2024 Findings.

Zimu Wang (2023–present, UT Dallas Visiting student) → UoL PhD.
Topic: Temporal relation extraction.
Publications: NeurIPS 2024, EMNLP 2024.

Son Tran (2023, UT Dallas Visiting student) → Cornell PhD.
Topic: Complex Question Answering.
Publications: NeurIPS 2024.

Chenkai Ma (2023, UT Dallas Visiting student).
Topic: Multiple Choice Questions Reasoning.
Publications: EMNLP 2023.

Barry Wang (2021–2024, Cornell CS undergraduate student) → CMU PhD.
Topic: Automatic Error Analysis for Information Extraction.
Publications: ACL 2022, SciNLP 2022, EMNLP 2023, NeurIPS 2024.

Zonglin Yang (2020–2024, Cornell CS MEng student) → NTU PhD.
Topic: Commonsense and Case-based Reasoning for NLP.
Publications: EMNLP 2020 Findings, EACL 2023, ACL 2024.

Ehsan Aghazadeh (2024, UT Dallas Visiting student).
Topic: Large Vision Language Models Interpretability.

Bowen Yan (2024, UT Dallas Visiting student).
Topic: Large Vision Language Models Hallucinations.

Zhengsong Zhang (2024, UT Dallas Visiting student).
Topic: Large Vision Language Models Hallucinations.

Ziming Luo (2024, UT Dallas Visiting student).
Topic: Large Language Model for Scientific Discovery.

Undergraduate/High School Students

Teerth Patel (UT Dallas BS student, 2023–present).
Topic: Large Language Models Peer Evaluations.
Publications: TMLR 2024.
Award: Jonsson School of Engineering and Computer Science Award.

Lei Xia (2023–2024, UT Dallas Visiting student).
Topic: Multi-hop Question Answering.
Publications: NeurIPS 2024.

Minhao Zou (2024, UT Dallas Visiting student).
Topic: LLM for Scientific Discovery.

Arjun Junghare (2024, UT Dallas CS undergraduate student).
Topic: Image Generation with Large Language Models.

Shreya Kumara (2024, President of UTD Society of Asian Scientists and Engineers).

Anjana Bharadwaj (2024, President of UTD Women Who Compute).

Jaden Nunes (Summer 2023, DFW Local K-12 student).
Topic: Event Extraction as Question-Answer Pairs Generation.

Rishab Bhattacharya (Summer 2023, DFW Local K-12 student).
Topic: Event Extraction as Question-Answer Pairs Generation.

Shreyas Kumar (Summer 2023, DFW Local K-12 student).
Topic: Event Extraction as Question-Answer Pairs Generation.

Aliva Das (2021–2022, Cornell CS undergraduate student) → Amazon.
Topic: Automatic Error Analysis for Information Extraction.
Publications: ACL 2022, SciNLP 2021.

Rishi Malhotra (Spring 2021, Cornell CS undergraduate student) → Microsoft.
Topic: Applying Neural Document-level IE Model to the Scientific Domain.

Maitreyi Chatterjee (Spring 2021, Cornell CS undergraduate student) → LinkedIn.
Topic: Applying Neural Document-level IE Model to the Scientific Domain.

Teaching Experience

Introduction to Machine Learning, UT Dallas, Spring 2024.

Natural Language Processing, UT Dallas, Fall 2023, 2024, 2025.

Deep Learning for Natural Language Processing, UT Dallas, Spring 2023.
New course developed and taught (overall Instructor Score of 4.75/5.0). [Link]

Natural Language Processing, UT Dallas, Fall 2022.

Natural Language Processing, Cornell University, Fall 2019.
Teaching Assistant for Prof. Claire Cardie.

Natural Language Processing, Cornell University, Spring 2019.
Teaching Assistant for Prof. Yoav Artzi.

Software Engineering, Cornell University, Spring 2017, 2018.
Teaching Assistant for Prof. William Arms.

Introduction to Computing Using Python, Cornell University, Fall 2016.
Teaching Assistant for Prof. Walker White.

Professional Services

Grant-Proposal Review Panelists

NSF Panel in 2026 (for IIS/RI Future CORe program).

NSF Panel in 2025 (for IIS/RI MEDIUM program).

Chairing Roles:

ICLR (International Conference on Learning Representations) Area Chair, 2025, 2027.

NeurIPS Area Chair, 2026.

ACL Demo Track Area Chair, 2026.

ACL (Annual Meeting of the Association for Computational Linguistics) Rolling Review Area Chair, 2024.

ACL Area Chair, 2023.

ACL Session Chair, 2024.

EMNLP (Conference on Empirical Methods in Natural Language Processing) Area Chair, 2024. **(Outstanding Area Chair Award).**

EMNLP Demo Track Area Chair, 2024.

NAACL (Conference of the North American Chapter of the Association for Computational Linguistics) Website Chair, 2024.

COLING (International Conference on Computational Linguistics) Senior Area Chair, 2024, 2025.

Seminar/Conference/Workshop/Tutorial Organizing:

Frontiers of LLMs for Automated Scientific Discovery: A Tutorial
Organizing team: Zonglin Yang, Chandan Reddy, Xinya Du and Mengdi Wang.
In AAAI Conference on Artificial Intelligence (AAAI), 2026

AI4Research: Towards a Unified Knowledge-grounded Scientific Research Lifecycle.
Organizing team: Qingyun Wang, Wenpeng Yin, Lifu Huang, Yi R. Fung, Xinya Du, Carl Edwards, Tom Hope.
In AAAI Conference on Artificial Intelligence (AAAI), 2025.

Evaluations and Benchmarks in Context of Multimodal LLM.
Organizing team: Hao Fei, Xiang Yue, Kaipeng Zhang, Long Chen, Jian Li, Xinya Du.
In The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.

The First Workshop on Multimodal Knowledge and Language Modeling
Organizing team: Liqiang Jing, Xinya Du, Hao Fei, Jing Gu, Manling Li, Aixin Sun,

William Wang
In International Joint Conference on Artificial Intelligence (IJCAI), 2025.

Journal Reviewer:

TKDE (IEEE Transactions on Knowledge and Data Engineering).

TASLP (IEEE Transactions on Audio, Speech and Language Processing).

TNNLS (IEEE Transactions on Neural Networks and Learning Systems).

TALLIP (ACM Transactions on Asian and Low-Resource Language Information Processing).

TKDD (ACM Transactions on Knowledge Discovery from Data).

CL (Computational Linguistics).

KAIS (Knowledge and Information Systems).

AI Communication.

IPM (Information Processing and Management).

Conference Committee Member:

ACL (Annual Meeting of the Association for Computational Linguistics).

NeurIPS (Conference on Neural Information Processing Systems).

ICLR (International Conference on Learning Representations).

AISTATS (International Conference on Artificial Intelligence and Statistics).

EMNLP (Conference on Empirical Methods in Natural Language Processing).

NAACL (Conference of the North American Chapter of the Association for Computational Linguistics).

AACL/IJCNLP (Asia-Pacific Chapter of the Association for Computational Linguistics / International Joint Conference on Natural Language Processing).

IJCAI (International Joint Conference on Artificial Intelligence).

AAAI (AAAI Conference on Artificial Intelligence).

NLPCC (Natural Language Processing and Chinese Computing).

CoNLL (Conference on Computational Natural Language Learning).

WNUT (Workshop on Noisy User-generated Text).

MRQA (Workshop on Machine Reading for Question Answering).

SEM (Joint Conference on Lexical and Computational Semantics).

PhD/MS Committee Member:

PhD: Kangshuo Li, Souradeep Nanda, Ali Riahi, Xiangci Li, Jishnu Jaykumar Padalunkal, Basel Abdeen, Yibo Hu.

MS: Wooseong Yang, Shubham Patel.

Other Activities:

Faculty advisor and judge: Association for Computing Machinery (ACM) Symposium, UT Dallas, 2023–present.

Faculty Advisor: Women Who Compute (WWC), UT Dallas, 2024–present.

Faculty Advisor: Society of Asian Scientists and Engineers (SASE), UT Dallas, 2024 – present.

Invited Speaker: UTD AI Society's "AI Chats", 2025.

Panelist: FuturED 2024 Workshop on Future of Event Detection, 2024.

Faculty Host: K-12 Outreach, UT Dallas, Summer 2023, 2024.

Research Mentor: Research, Inquiry, Design Experience (RIDE) Project, UT Dallas, 2024.

PhD Admission Committee Member, UT Dallas CS Department, 2022, 2023, 2024.

Site Host: North American Computational Linguistics Olympiad (NACLO), 2023, 2024.

PhD Admission Committee Member, Cornell CS Department, 2021.

Volunteer: Cornell CS Department PhD Visit Day, 2019, 2020, 2021.

Student Volunteer: ACL, 2017, 2018.

Student Volunteer: EMNLP, 2017.

Invited Talks

Pushing the Boundaries of LLM Agents in Data Science and Machine Learning Research
Nvidia, Dec 2025.

Inside LLMs: Knowledge, Alignment, and Agents.
Samsung Research America (SRA), May 2025. .

Towards Autonomous Scientific Discovery with Large Language Models.
UT Southwestern AI in Medicine Seminar Series, May 2025.

Synergizing Knowledge and Large Language Models.
University of Illinois Urbana-Champaign, Data Mining Group Seminar, Oct 2024.
University of Massachusetts–Amherst, Machine Learning & Friends Lunch, Sep 2024.

Synergy between Large Language Model and Knowledge.
University of North Texas, Oct 2024.

Panel at Future of Event Detection Workshop.
Conference on Empirical Methods in Natural Language Processing (EMNLP), Nov 2024.

Faculty Round Table Talk.
UT Dallas Hobson Wildenthal Honors College, Aug 2024.

Large Language Models: Knowledge, Reasoning and Factuality.
Samsung Electronics America, Mar 2024.

Making Natural Language Reasoning Explainable and Faithful.
AAAI Conference on Artificial Intelligence (AAAI), Jan 2024.

Open-ended Evaluations of Foundational Models: Alignment and Faithfulness.
Shanghai Jiao Tong University, Computer Science Department Seminar, Dec 2023.
Fudan University, Computer Science Department Seminar, Dec 2023.

ChatGPT: Fact vs. Fiction.
UT Dallas, Forum sponsored by The Dallas Morning News, May 2023.

Towards More Intelligent Extraction of Information from Documents.
Allen Institute for Artificial Intelligence, Mar 2022.
Facebook, Mar 2022.
Simon Fraser University, Feb 2022.

Rensselaer Polytechnic Institute, Feb 2022.
University of California, Merced, Feb 2022.

Towards More Intelligent Extraction of Information from Documents.
UIUC, Siebel School of Computing and Data Science Speaker Series, Feb 2022.

Towards More Informed Extraction of Events from Documents.
University of Chicago, Rising Stars in Data Science Workshop, Jan 2021.
Tencent AI Research America, Nov 2020.

Event Extraction by Answering (Almost) Natural Question.
UIUC, Information Extraction and Knowledge Acquisition Class, Sep 2020.

LwLL: Progress on the NLP Front.
Cornell University, DARPA site visit, Apr 2020.

THE UNIVERSITY OF TEXAS AT DALLAS

800 W. Campbell Road EC-31 RICHARDSON, TEXAS 75080-3021
Phone: 972-883-2091 Fax: 972-883-2399

COMPUTER SCIENCE DEPARTMENT

ERIK JONSSON SCHOOL OF
ENGINEERING AND COMPUTER SCIENCE

October 20, 2026

Dear ONR Young Investigator Program Selection Committee:

I am pleased to provide my strongest endorsement for Dr. Xinya Du's application to the Office of Naval Research Young Investigator Program, entitled "Trustworthy Collective Reasoning and Evaluation for AI Agents." As an Assistant Professor in the Department of Computer Science at The University of Texas at Dallas, Dr. Du is an outstanding early-career researcher with exceptional potential to become a leader in artificial intelligence, with a focus on language model reasoning, evaluation, and autonomous agents. Our department is strongly committed to supporting his research and long-term academic career.

Dr. Du was recruited to UT Dallas based on his strong research accomplishments, technical expertise, and ambitious vision for advancing artificial intelligence. His recruitment reflects our department's commitment to attracting outstanding researchers with the potential to make significant contributions to their fields. Since joining the department, he has established an active research program addressing fundamental challenges in the reliability and reasoning capabilities of modern AI systems. His research has resulted in publications at leading conferences in natural language processing and machine learning, including NeurIPS, ACL and EMNLP. His ability to identify important emerging problems and develop original technical solutions distinguishes him as a highly promising young investigator.

Dr. Du's proposed ONR research, "Trustworthy Collective Reasoning and Evaluation for AI Agents," addresses a fundamental challenge in the development of reliable autonomous AI systems: enabling multiple agents to evaluate one another's reasoning, recognize relevant expertise, and improve collective decisions even when individual agents are imperfect or unreliable. The project seeks to develop novel methods for peer evaluation, collective reasoning, and trustworthy collaboration among AI agents. This research has significant potential to advance the foundations of multi-agent intelligence and contribute to reliable autonomous decision-making in complex environments. Dr. Du's expertise in language model reasoning and evaluation, together with his ongoing work on AI agents, positions him well to pursue this ambitious research agenda.

The University of Texas at Dallas is committed to providing sustained institutional support for Dr. Du's research and career development. The department has provided him with start-up funding, dedicated research space, and access to computing resources. We also support his research through a reduced teaching workload, support for graduate student recruitment, faculty mentoring, and professional development opportunities. These institutional investments will help Dr. Du further develop his independent research program and expand its impact. Our commitment extends beyond the proposed ONR award, reflecting our confidence in his long-term potential and our investment in his development as a leading faculty member.

Dr. Du exemplifies the qualities sought by the ONR Young Investigator Program: scientific creativity, technical excellence, research independence, and exceptional promise for sustained contributions to his

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field. I strongly endorse his application and reaffirm our department's commitment to supporting his continued success at The University of Texas at Dallas.

Sincerely,

Rym Zalila-Wenkstern



Rym Zalila-Wenkstern, Ph.D.
Professor and Interim Department Head
Department of Computer Science
The University of Texas at Dallas
Email: rymw@utdallas.edu
Phone: (972) 883 2091

9 Data Management Plan

Types of data, software, and other materials. The project will produce (1) three testbeds of questions with known answers: math and code problems with planted shared mistakes (A), multi-document questions composed from event structures, including synthetic maritime scenarios (B), and scientific-claim and peer-review questions with planted errors (C); (2) logs of agent answers, ratings, debates, tool calls and their outputs, judge rulings, and audit outcomes; (3) software: the agentic debate harness, the shared-error-factor rater model, the impartial selection mechanism, the audit scheme, red-team agents, reinforcement-learning code, and the simulation code behind the theoretical results; and (4) fine-tuned weights of open-weight models, where the original licenses permit release.

How the data will be acquired. Testbed A draws on public benchmarks (GSM8K and MATH, released under the MIT license) and public programming problems. Testbed B is generated by our MEQA pipeline from event structures; its maritime scenarios are fully synthetic and unclassified. Testbed C uses public papers and the public reviews and rebuttals posted on OpenReview, under the platform's terms, together with published reproducibility reports. Logs are produced by running agents built on open-weight models on UT Dallas computers and on commercial models through their interfaces. No human-subjects data are collected; in experiments, human audits are simulated by revealing known answers.

Time and location of acquisition. Not scientifically pertinent: all data are generated or collected computationally at UT Dallas during the 36-month project.

Processing. Questions are normalized to a common schema, deduplicated, and split into development, evaluation, and held-out sets by topic and by model family. Planted cases are generated by versioned scripts that record each edit and its effect on the answer. Logs are stored as produced and then summarized for analysis; every number in a publication is regenerated by a released script from the stored logs.

File formats and naming conventions. Questions and logs are stored as JSON Lines and Parquet; code is Python, released under the Mozilla Public License 2.0, as in the PI's earlier projects; model weights use the safetensors format. File names give the testbed, split, and version (for example, `tbB_eval_v1.2.jsonl`), with semantic versioning and a change log for each release.

Quality assurance and quality control. The known answer of every generated question is confirmed by construction and by an independent script; two students review a random sample of each release, and disagreements are resolved before release. Each log record carries the model version, prompt version, random seed, and a hash of tool outputs, so that any debate can be re-run. The harness is frozen at each milestone, and automated runs confirm that frozen versions reproduce stored results.

Origin of existing data. GSM8K (Cobbe et al., 2021), MATH (Hendrycks et al., 2021), the MEQA benchmark and pipeline (Li et al., 2024b), public OpenReview reviews, published reproducibility reports, and PeerEval logs produced by the PI's group. Each release records the source and license of every component.

Standards for data and metadata. Datasets will carry Croissant machine-learning metadata and a datasheet describing their purpose, composition, collection, and known limitations; code repositories will include environment files that pin software versions.

Preservation and access. Code will be public on GitHub, with each milestone release archived on Zenodo under a DOI. Testbeds and logs will be released on Hugging Face and on project webpages linked from the PI's lab website, by the time of the associated publication, and kept available for at least five years after the project ends. Working copies stay on access-controlled lab machines, with secondary copies on departmental systems at least once per semester; HPC scratch storage is never used as an archive. Final reports will be submitted as required by DoD Instruction 3200.12.

Balance of value and cost. We will keep all data behind published results. Raw logs of exploratory runs,

which can be large, will be kept for one year and then reduced to summaries, since the released code can regenerate them.

Security. All data are public or synthetic, unclassified, and contain no controlled unclassified information or personally identifiable information. If any data were later found to raise national security concerns, they would not be released, in accordance with DoD Instruction 5230.09.

References Cited

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J. Biswas et al. (2026). AI-Assisted Peer Review at Scale: The AAAI-26 AI Review Pilot. arXiv:2604.13940.

S. R. Bowman et al. (2022). Measuring Progress on Scalable Oversight for Large Language Models. arXiv:2211.03540.

J. Brown-Cohen, G. Irving, and G. Piliouras (2024). Scalable AI Safety via Doubly-Efficient Debate. In Proceedings of the 41st International Conference on Machine Learning (ICML 2024), PMLR 235:4585–4602.

J. Brown-Cohen, G. Irving, G. Piliouras, L. Chen, J. Li, and Z. Xun (2025). Avoiding Obfuscation with Prover-Estimator Debate. arXiv preprint arXiv:2506.13609.

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Proceedings of the 22nd International World Wide Web Conference (WWW '13), pp. 319–330.

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Y. Denisov-Blanch, J. Kazdan, J. Chudnovsky, R. Schaeffer, S. Guan, S. Adeshina, and S. Koyejo (2026). Consensus is Not Verification: Why Crowd Wisdom Strategies Fail for LLM Truthfulness. arXiv:2603.06612.

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X. Du et al. (2022). RESIN-11: Schema-guided Event Prediction for 11 Newsworthy Scenarios. NAACL-HLT: System Demonstrations, pp. 54–63.

Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch (2024). Improving Factuality and Reasoning in Language Models through Multiagent Debate. ICML, PMLR 235:11733–11763.

S. Eo, H. Moon, E. H. Zi, C. Park, and H. Lim (2025). Debate Only When Necessary: Adaptive Multiagent Collaboration for Efficient LLM Reasoning. arXiv:2504.05047.

W. Fan, J. Yoon, and B. Ji (2026). iMAD: Intelligent Multi-Agent Debate for Efficient and Accurate LLM Inference. AAAI.

B. Flanigan, P. Gölz, A. Gupta, B. Hennig, and A. D. Procaccia (2021). Fair Algorithms for Selecting Citizens' Assemblies. Nature 596(7873):548–552.

J. S. Gans and R. T. Holden (2026). When Does Randomized Oversight Align AI Agents That Can Conceal? NBER Working Paper 35878.

S. Goel, J. Strüber, I. A. Auzina, K. K. Chandra, P. Kumaraguru, D. Kiela, A. Prabhu, M. Bethge, and J. Geiping (2025). Great Models Think Alike and this Undermines AI Oversight. ICML, PMLR 267:19621–19678.

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B. He, Y. Zuo, Z. Liu, S. Zhao, Z. Fu, J. Yang, C. Qian, K. Zhang, Y. Fan, G. Cui, et al. (2026). How Far Can Unsupervised RLVR Scale LLM Training? ICLR, pp. 14823–14865.

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J. Hu, A. Shen, and K. Lakshmipathi (2026b). Statistical Scouting Finds Debate-Safe but Not Debate-Useful Cases: A Matched-Ceiling Study of Open-Weight LLM Reasoning Protocols. arXiv:2605.09618.

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G. Irving, P. Christiano, and D. Amodei (2018). AI Safety via Debate. arXiv:1805.00899.

A. Jaffe, E. Fetaya, B. Nadler, T. Jiang, and Y. Kluger (2016). Unsupervised Ensemble Learning with Dependent Classifiers. AISTATS, PMLR 51:351–360.

S. Jeong, Y. Choi, J. Kim, and B. Jang (2026). Tool-MAD: A Multi-Agent Debate Framework for Fact Verification with Diverse Tool Augmentation and Adaptive Retrieval. arXiv:2601.04742.

L. Jing, R. Li, Y. Chen, and X. Du (2024). FaithScore: Fine-grained Evaluations of Hallucinations in Large Vision-Language Models. Findings of EMNLP 2024, pp. 5042–5063.

A. Kahng, Y. Kotturi, C. Kulkarni, D. Kurokawa, and A. D. Procaccia (2018). Ranking Wily People Who Rank Each Other. AAAI, pp. 1087–1094.

D. Kang, W. Ammar, B. Dalvi, M. van Zuylen, S. Kohlmeier, E. Hovy, and R. Schwartz (2018). A Dataset of Peer Reviews (PeerRead): Collection, Insights and NLP Applications. NAACL-HLT (Volume 1: Long Papers), pp. 1647–1661.

Z. Kenton et al. (2024). On Scalable Oversight with Weak LLMs Judging Strong LLMs. NeurIPS 37, pp. 75229–75276.

Z. Kenton, L. Janzer, R. Greig, T. H. Teh, K. Tyshchuk, J. Brown-Cohen, H. Edwards, S. Rajamanoharan, N. Y. Siegel, N. Jaques, and R. Shah (2026). Debate Training Reduces Reward Hacking in RLAIF. arXiv:2608.17776.

A. Khan et al. (2024). Debating with More Persuasive LLMs Leads to More Truthful Answers. ICML, PMLR 235:23662–23733.

E. M. Kim, A. Garg, K. Peng, and N. Garg (2025). Correlated Errors in Large Language Models. ICML, PMLR 267:30038–30066.

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